The Accessible "ndsu-thesis-tagged" LaTeX class—Documentation

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GitHub: `https://github.com/CIgathi/NDSU-Thesis-Class.git` (link:)

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1 Features of the NDSU LaTeX Tagged PDF Template

Features of the NDSU LaTeX Tagged PDF Class

- PDF tags are automatically created
- Produces the latest PDF format standard
- Automatic PDF document properties
- User-friendly tamplate design
- Supports tables, figs., schemes, and listings
- Supports 5 disquisitions and 2 degree types
- Both full-justified and left-justified theses
- Supports several fonts types (28)
- Automatic filler text and sample figures
- Automatic headings formats and tags
- Easy landscape figures and tables
- Alternate text added with captions
- Automatic bibliography formats
- Multiple named appendices
- Line numbering
- Supports TikZ vectorized drawings
- Several handy shortcuts were defined
- Near-perfect accessibility score ($\geq$ 98)
- User concentrate on content LaTeX formats
- Users input their LaTeX content as usual
- Extended math support included (MathML)
- Automatic PDF bookmarks
- Examples given to cover most situations
- All these elements in supported in appendix
- Both numbered and non-numbered theses
- Showgrid option for vertical positioning
- Makes paper-style & chapter-style outputs
- Automatic TOC and other lists
- Automatic full-width tables and longtables
- Easy listings of source codes
- Cite-while-you-write methodology
- Individual chapter or combined bibliography
- Block comments and temporary ending
- Annotation commands for review work
- Supports chemical symbols and reactions
- Easy conversion—thesis to journal articles
- Class updated regularly based on demands
- All advantages of LaTeX come for a ride!

2 How to Use the Template—Step-By-Step Procedure

First of all, it is easy to use the template for making a dissertation (or LaTeX any other project for that matter). It is assumed that user have a LaTeX in their system running (desktop version is preferred to online version—more on this later). Just follow these steps:

Step 1. Download project, extract, and have a copy: Download the zipped project file ("`NDSU-tagged-thesis-GitHub.zip`") from the GitHub repository (link:) and extract the contents. You can either retain the zip file for extraction if necessary or have a copy of the extracted folder.

Step 2. Know the source: The project contents will have (1) "`ndsu-example-tagged.tex`" (source code file that users will work), (2) "`mybib.bib`" (bibliography source file that users will update), (3) "`ndsu-example-tagged.cls`" (class file that users should not modify and is used by the source code), (4) "`ndsu-example-tagged.pdf`" (output tagged PDF file generated from the source code for users to see), (5) "`figures`" folder (where users can store their images [graphs, pictures, schemes, etc.] conveniently, and the source code will fetch from this folder), (6) "`...documentation....pdf`" (documentation of the class file ["this file"] that users must read for efficient use of this class and generation of dissertation), (7) "`barebone-ndsu-example-tagged.tex`" (fully-functional source code of the dissertation that users can try on small chunks that can be transferred to the main source file), and (8) "`barebone-ndsu-example-tagged.pdf`" (output tagged PDF file generated from barebone source code). Initially, the list of project files may appear overwhelming, but they are readily available and become comprehensible as we proceed. It is just enough to have files (1)–(3) to generate the output.

Step 3. Open the source and run: Open the *.tex source file (1) using the desktop LaTeX software installed in your system and compile (run) to regenerate the output (4; fig. 1 on page 8). That is all there is to it. In LaTeX terminology, running means "compiling" the document enough number of times. A general run for tagged PDF involves compiling using LuaTeX (or LuaLaTeX) + Biber + LuaTeX + LuaTeX for documents with a bibliography. It is possible to set the preferences to make these four compilations in a single command. It is important to note that a single LuaTeX compilation will generate the output, but cross-references and citations require additional compilations.

Step 4. Add your content: Users can now modify the contents of the example source code and see the results after compilation. Typically, users replace the text of the arguments {...} (e.g., `\title{...}`, `\author{...}`, `\keywords{...}`, etc.) without altering the commands (thosepreceded by \(e.g., `\title`, `\abstract`, `\listofabbreviations`, and so on). As necessary, the existing text can be replaced or new (both commands and argument text) can be added (e.g., new sections, subsections, chapters, floats, appendices, etc.), based on the content of the dissertation at hand.

Step 4. When things fail: As the project was thoroughly tested before the release, there were no issues encountered. Common causes of compilation failure are (i) incomplete installation of LaTeX, (ii) LaTeX installation and packages are not updated, and (iii) use "`biber`" not "`BibTeX`"—bibliography compilers for tagged PDFs. Utilize the TeXLive (updating utility) to update the packages and restart the machine to ensure the update takes effect.

Step 5: General thoughts It should be understood and recognized that LaTeX is highly advanced, open-source, free, well-supported by a knowledgeable users community, and is capable of generating very large complex documents, such as mathematics books, technical journals, and large publication projects, and the graduate dissertation tried to be created pales in comparison. It is never the system of LaTeX to be blamed, but the bugs we introduce along the way, unwittingly. Fortunately, every possible error has a solution, and finding the solutions is easy nowadays with the tools available to us (e.g., stock exchange, AI). It was widely observed that students who utilized LaTeX consistently enjoyed the experience (never looked back—solely because of the quality output) and employed it throughout their professional careers.

3 Accessible NDSU Thesis using LaTeX Template (Tagged PDF)

An accessible PDF document, simply put, is a tagged PDF document that has additional instructions to be used by assistive technology (AT) to help individuals with disabilities or those who require assistance. Related to PDF documents, examples of AT include screen readers and braille embossers that support inclusivity and make the document available to all, also becoming a DOJ legal requirement. All NDSU dissertations should adhere to this paradigm (WCAG 2.1 AA compliance requirements), which means untagged PDF dissertations will no longer be accepted after April 2027. Therefore, the development of this new `ndsu-thesis-tagged` LaTeX class is necessary, which henceforth contains the required codes to fulfill additional PDF requirements.

Several NDSU resources can be accessed (link: NDSU's resources on Accessibility and Universal Design) to gain knowledge about the subject. The original LaTeX Tagging Project home page (link: LaTeX Tagging Project) contains all the LaTeX related resources on this topic. Students and users utilizing this `ndsu-thesis-tagged` class (NTTC) should be aware of the information presented within the colored textbox. This awareness will demystify the entire concept of generating tagged PDF dissertations. However, for some of the concepts presented, a knowledge of LaTeX and dissertation writing is required, which can be acquired by starting from the Introduction (Section 5).

NDSU LaTeX Tagged (PDF 2.0 + /UA-2) PDF Dissertation: Important Information

As everyone is learning and adapting to the new paradigm of accessible documents, additional information will be periodically added. Users are encouraged to visit this document for the most up-to-date information. Recently, LaTeX updated the packages [`latex-lab-testphase-block v1.0a` (2026-05-28)] and this changed all the chapter (H1) heading styles, hence needed a significant update in the class to render the final output similar to before the update state (Mid May, 2026).

- Students and users should start using the new "`ndsu-thesis-tagged`" LaTeX class (NTTC) and resouces deveoped (link:) and not the previous thesis class (`ndsu-thesis-2022`)

- Tagged PDF looks similar to a regular PDF, but contains additional instructions for AT, hence the file size will be much larger (e.g., 7.5 times!)
- As LaTeX is already a structured language, it lends itself well to the creation of accessible documents (tagged PDF)—meaning, "we are already there—and nothing much needs to be done!"
- Using NTTC, the "Document Properties" from Adobe Reader will show: "Description > Tagged PDF: Yes," (other items: Title, Author, and Keywords filled) and "Advanced > Language: English"
- Use of NTTC also automatically populates the PDF bookmarks, including the "TITLE PAGE" bookmark and chapters up to the level of subsection as available in the TABLE OF CONTENTS
- Using NTTC automatically tags most elements, such as heading levels (H0, H1, H2, H3, H4, ...), tables (caption, TH and TD), figures (caption, alt text), and bookmarks

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- The previous NDSU LaTeX thesis class (`ndsu-thesis-2022`; which is "untagged PDF") will work almost "out-of-the-box," provided the minor modifications (preamble `DocumentMetadata`)
- Any LaTeX document can be readily converted into an accessible document (tagged PDF) by adding the `\DocumentMetadata{...}` as the very first command/instruction
- This command `\DocumentMetadata{tagging=on,lang=en-US,tagging-setup={math/setup=mathml-SE, math/alt/use},pdfversion=2.0,pdfstandard=ua-2,pdfstandard=a-4f,uncompress}` represents the latest and advanced tagging arrangement, including advanced math support
- A new `\keywords{...}` command, usually after the `\title{...}` and `\author{...}`, in NTTC will populate the PDF's keywords property, while the title, author, and language properties were automatically filled
- Modifications include replacing the commands from "incompatible" packages by simpler commands or from compatible packages (link: LaTeX Tagging Status)
- It is okay to use some of the "partially-compatible" packages (e.g., `siunix`) as long as the commands used compile and output generated as expected
- With NTTC, while compiling, the LuaTeX engine should be used as it provides superior tagging capabilities (LuaTeX + Biber + LuaTeX + LuaTeX; for documents with bibliography)
- Some of the useful steps to make it run—sometimes LuaTeX, switching from pdfLaTeX, may not run the first time: (i) Update LaTeX packages, (ii) Restart the machine for updates to take effect, and (iii) Ensure "`biber`" is run (not BibTeX) for the bibliography creation.
- Most of the supporting resources of the previous `ndsu-thesis-2022` will be applicable (presented again with modifications here), with relevant updates added subsequently

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- Previously highly useful packages, such as `threeparttable`—tablenotes env., `tabularray`—tblr env., `chemformula`, `listings`, etc. are currently incompatible with tagging
- Functionality of the aforementioned packages, in NTTC, can be coded using simpler commands or packages such as `\multicolumn{...}`, `xltabular`, `mhchem`, and `fancyverb` packages
- Some of the common symbols (`\qty{100}{\degreeCelsius}` = 100 °C) need to coded through as `\qty{100}{\textdegree{C}}` (simpler); and `\ch{H2SO4}` = H_2SO_4 as `\ce{H2SO4}` (compatible)
- The float placement specifier [H], comes from the `float` package, which is useful to force the placement, but it is not compatible anymore, and should be replaced by [h!]
- In tables, the command `\tagpdfsetup{table/header-rows={1}}% for tagging the header row; rows={1,2} for two rows`, before the tabular or xltabular env. tags the header rows
- In figures or schemes, the alternate text (which is different from the caption) can be coded in the optional argument of `\includegraphics[width=\textwidth, alt={...your alt text ...}]`; alternatively, using the `\mytfig`[vspace]{h!}{size}{file name}{caption}{alt text}{label} shortcut
- A family of shortcuts, such as `\mytfig`, `\mytfigls`, `\mytsch`, `\mytschls`, etc., for figures and schemes, both for regular and appendix, are created

- Simpler listing of codes is achieved through the new environment `mytlisting` and `\myliscap{...}` and `\mylisapcap{...}` for regular and appendix listing captions and prefatory lists generation

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- Accessibility of the dissertation can be checked through PDF Accessibility Checker (PAC; Windows), axesCheck (Mac), or Adobe Acrobat Pro (AAP) itself; but AAP is required for fixing tagging issues
- AAP (available through NDSU's "virtual lab" system and dedicated machines) should be used to perform the "Check for accessibility" and fix various accessibility-related issues
- Students are encouraged to utilize available resources and dedicated time to learning accessibility checking and fixing through AAP, as these concepts were not typically covered in class
- Since the LaTeX-generated dissertation is already tagged with proper reading order, the "fixes" performed through AAP are expected to be minimal (BB Ally checker often gives a perfect score)
- Things that need to be fixed in Ally checker are (i) "This PDF does not have a title" — this (is a common bug) can be fixed by inputting the title and "Apply fix," and (ii) "This PDF contains text with insufficient contrast" — this can be easily fixed by better text with contrast (text *vs* background; e.g., ABCD■ fails and ABCD■ passes (30% black added; link: Color checker)
- Thus, gone are the days when we randomly selected colors for the text. Now, the chosen text color should have a contrast ratio ≥ 4.5 w.r.t. background, hence the choice is restricted (refer to the link for some text with background color selection and their contrast ratio: Color checker)
- AAP serves as the ultimate tool for checking, fixing, and determining the accessibility of the PDF document; therefore, it is crucial for students to acquire proficiency in the AAP accessibility tools.

4 Frequently Made Mistakes (FMM)

Following the top-down approach, the following are the frequently made mistakes by the students, in other words—the most useful concepts, observed over the years, are presented to capture their immediate attention.

Most Useful Concepts or FMM

- Not reading the class documentation and resources (link:)
- Not leaving a blank line to create a new paragraph—using \\ instead
- A text paragraph should be a single unit—not made by several lines with hard return
- Proper use of vspace (+ve and -ve values) around headers, tables, and figures (Section 13.7)
- Use of non-ASCII or other format characters in source code (*.tex and *.bib; Section 13.11)
- Duplicate entries in bibliography
- Incomplete or too short bibliography entries (e.g., one-line entry)
- Inconsistent formatting of title (titlecase *vs* sentence case) and journal name (abbreviated *vs* full)
- Proper format of titlecase *vs* sentence case in various heading levels
- Not utilizing the excellent functionality of the SI units package
- Not taking advantage of several convenient shortcuts developed in the thesis class (e.g., H1, H2, H3, etc.)
- Not utilizing the block comment and temporary ending command to focus on the current work block
- Font size of the text in figures should be legible (e.g., about 80 % to 100 % of the regular text)
- Proper use of the landscape page to fit larger content—for tables and figures
- Not using the chemical typesetting packages, but using math mode to manually code chemicals
- Not following the appropriate convention in coding equations and treating the symbols in text
- Incorrect color contrast ratio for text and background (should be ≥ 4.5; see examples—link: Color checker)

5 Introduction

The NTTC is an updated version of the previous non-tagged version of the thesis class file. This class generates disquisitions intended to comply with the disquisition requirements of the North Dakota State

University (NDSU) Graduate School. This class simulates the output as generated by the NDSU disquisition templates updated March 2026 (link: NDSU Disquisition Templates). This class is not officially endorsed by NDSU or the NDSU Graduate School, but efforts are underway toward that goal. It should be noted that several theses and dissertations were made and got approved by the Graduate School using the NDSU LaTeX thesis class in the past. Since disquisition requirements are subject to change at any time, the user is advised that the most current disquisition style policies supersede this class.

Most Important Instruction

For better and successful use of this NDSU LaTeX thesis template, the users should:

- READ ALL RESOURCES—documentation, example, barebone, vertical spacing, & extended thesis
- Use the LATEST CLASS in the GitHub (link:)
- MOST issues encountered by users were identified, addressed, and updated through examples
- So, "*No reinventing the wheel*," just READ, and it may be enough; otherwise, REPORT issues.

However, following the Graduate School-approved templates and collected experience from several previously approved dissertations, this LaTeX class was coded to incorporate the various required features and lessons learned. To ensure compliance with all NDSU Graduate School requirements, the user is encouraged to consult the NDSU Graduate School webpage and the links provided for detailed requirements and guidance on disquisition formatting guidelines, templates, section formatting, and examples.

The bundled template or the thesis example given (section 7) can be used as an easy starting point for using the class. Modification of the class file's code may result in unexpected behavior and is at the user's own risk. We recommend including additional packages and commands in the source file (*.tex) itself for the desired customization, as required by the departments and the users.

6 Using and Installing LaTeX—Online and Desktop Environments

LaTeX consists of the base software and the integrated development environment (IDE) to conveniently work with document development. Base LaTeX software for different operating systems can be downloaded from this resource (link: The LaTeX Project). Details on the LaTeX Tagging Project, codes and packages, are found at the link: The LaTeX Tagged PDF Project. Several online (e.g., Overleaf, Kile LaTeX Editor, Authorea, Papeeria, and so on) and standalone desktop versions (e.g., TeXMaker, TeXWorks, TexShop, TeXStudio, Prism - AI LaTeX Editor, and so on) of LaTeX IDEs are available. Online editors are "ready-to-go," with several templates, tutorials, and help documentation, where the user does not install the software but requires an internet connection. The desktop version requires software installation and updating (usually required after initial installation and needed only to employ new features, but it will run with existing packages). With base LaTeX installed, it is possible to edit source code and compile using command line processing (e.g., cmd Window or Mac terminal).

It is always preferable to have a standalone LaTeX desktop version for bigger projects (e.g., theses, papers), as it eliminates any obstacles in compiling the documents. In this case, the user has complete control, and everything is free with the open-source movement. While the online systems, which operate solely through internet connectivity, may be good for simple documents, they may run into "compilation timeouts" for larger documents or result in users paying fees for continued service or purchasing a plan. This whole idea of paying a fee for LaTeX is against the core principle of the developers of LaTeX (Knuth, D. and Lamport, L.), who made the system free for all to use in the first place. Therefore, the recommended system that offers the best user control, does not require internet connectivity, scalable documents, cost-free, and open-source is any standalone LaTeX desktop version.

Resources (text and video instructions) are available on how to use the online editor and install the LaTeX desktop version of the users' choice. As LaTeX is open source, most of these IDEs/editors are free, and usually, it is not necessary to spend on paid services to work with LaTeX and generate a document like a graduate disquisition or journal articles. It should be noted that LaTeX (manual released in 1986) is more than 40 years old, and LaTeX continues to be used as a high-quality free document preparation system by users of STEM fields.

7 Thesis Example—The Sample Template

7.1 The sample template

With LaTeX, the users type some commands and texts that are specific to their thesis/dissertation, which is human-readable (source code), as shown below (section 7.3). When the template that uses the developed NTTC was filled (with details and content—source code), and compiled, it will automatically generate the well-formatted NDSU thesis-style document that is automatically tagged (fig. 1).

The benefits of using LaTeX for thesis/dissertation include overall automation, open-source, freely available, vibrant society support, professional quality outcome, elegant mathematics handling, automatic bibliography management, integrated typography principles, portability among operating systems, longer life of the source code, every aspect of document preparation addressed, packages available for specialized needs, thesis/dissertation source code easily converted to journal articles with appropriate templates, and so on.

7.2 Packages already loaded in the class

The "`ndsu-thesis-2026 LaTeX class`" has been loaded with several commands, shortcuts, and supporting documents. The latest version can be obtained from the GitHub page (link:). It is recommended to download the latest resources by visiting this GitHub page before starting the thesis project. The following list of packages was already loaded into the class (users need not reload them):

- **Compatible LaTeX PDF tagging packages:** `booktabs, comment, etoolbox, [T2A,T1] fontenc, [stable] footmisc, ifthen, indentfirst, [utf8] inputenc, kantlipsum, longtable, mhchem, mwe, pdflscape, rotating, setspace, suffix, tabularx, textcomp, [nottoc] tocbibind, [figure, table] totalcount, [dvipsnames*,svgnames,x11names,table] xcolor, xltabular, xspace, & xurl.`
- **Partially compatible LaTeX PDF tagging packages:** `afterpage, [nameinlink] cleveref, csquotes, enumitem, [linktocpage,breaklinks,linktoc=all,hidelinks] hyperref, lineno, siunitx, tikz, [colorinlistoftodos,prependcaption,textsize=scriptsize] todonotes, & wasysym.`
- **Incompatible LaTeX PDF tagging packages:** `[export] adjustbox, amsfonts, amsmath, [labelsep=period] caption, chemformula, float, listings, [within=chapter] newfloat, subfig, tabularray, threeparttable, threeparttablex, titlesec, titling, [titles] tocloft, [normalem] ulem, & xtab.`

In view of the current standards on developing "Accessible Documents," where all NDSU theses are to be compliant (April 2027 onwards [1 year waiver from the previous decision]). Therefore, it is advisable to use the compatible packages predominantly and partially compatible packages with caution by observing the outputs. More on this will be made available in the near future. Furthermore, the following libraries are also loaded in the class:`usetikzlibrary{matrix, arrows, decorations.pathmorphing}`

It is expected that the "Compatible LaTeX PDF tagging packages" play well with the accessible tagged PDF creation (link: Tagged PDF). As this LaTeX PDF tagging project is ongoing, more packages will be worked on and added to the compatible list (1823 packages analyzed, and 858 compatible as of June 2026). Updates to NTTC will be applied as more compatible packages are developed and added to the core LaTeX repository.

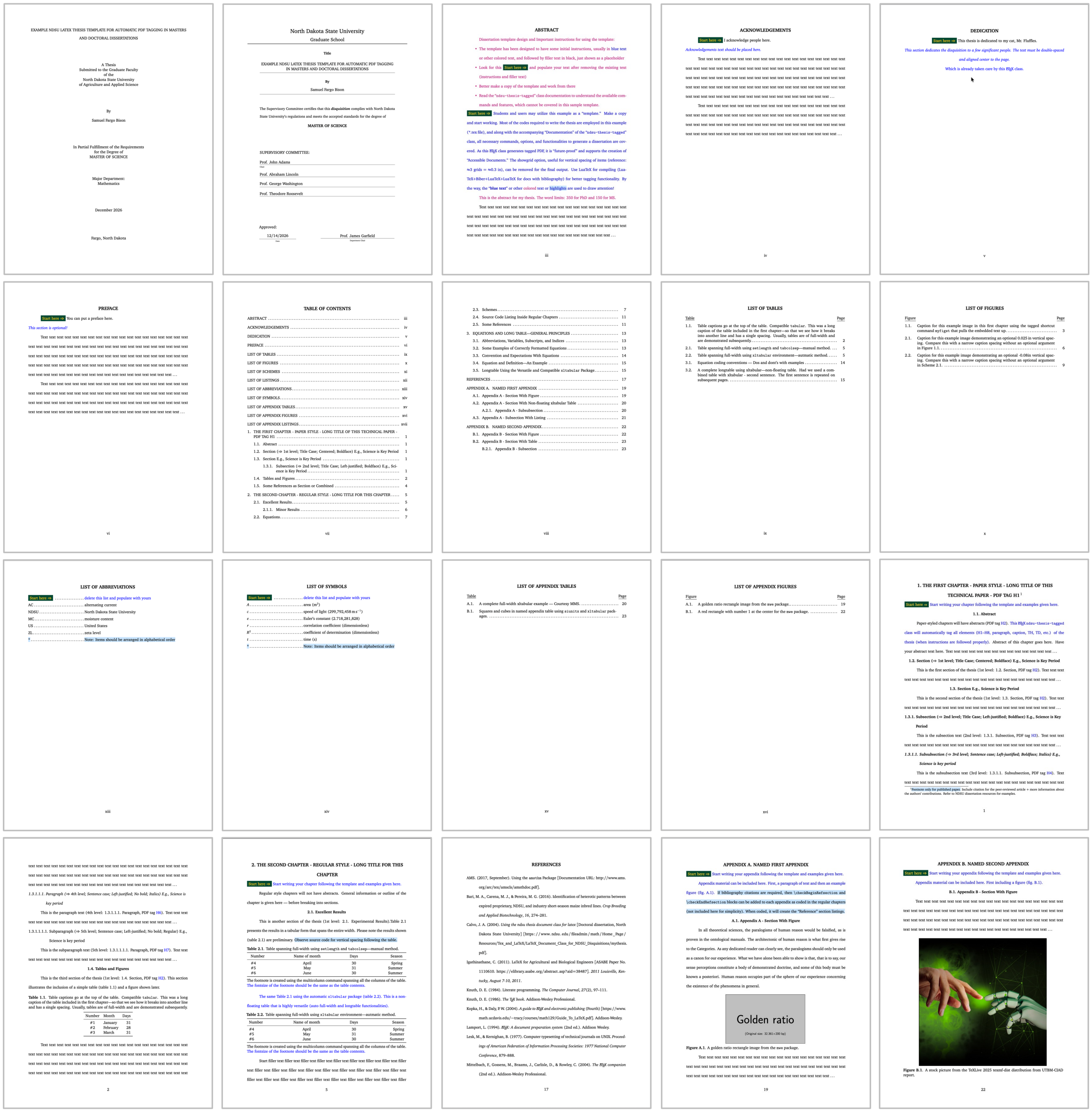


Figure 1: Automatically formatted output sample pages of the example thesis according to NDSU Graduate School requirements. Several pages were skipped to show the overall outcome and the source code.

7.3 "ndsu-example.tex"—the example thesis template

Presented in this section is a brief example of an M.S. thesis (copy/load code in the editor, have necessary resources [class + figures], and compile for output) that includes all required and several optional elements. An attempt was made to cover most of the aspects (prefatory items, chapters, sections, tables, figures, appendices, etc.) encountered during the preparation of the disquisition using LaTeX, therefore, the example is relatively elaborate. This example M.S. thesis code shown (link:) is included in the file named "`ndsu-example-tagged.tex`". In this example, the examining committee includes the Committee Chair, no Co-Chairs, and only two additional Committee Members. For this example, a "bibliography" file named `mybib.bib` was included separately, and BibLaTeX bibliography management system (advanced, versatile, and produces individual chapter bibliography listings) was used to manage references. The example thesis was optimized for the `mathdesign`, and with other fonts, some spacing adjustments should be made. However, this process is natural when a user develops the thesis section by section, following the ideas described in the example thesis template.

```
%******************* Metadata for PDF tagging *** [Updated: 10-06-2026] *******************
\DocumentMetadata{
  tagging=on,
  lang=en-US,
  tagging-setup={math/setup=mathml-SE,math/alt/use},
  pdfversion=2.0,
  pdfstandard=ua-2,% ua-1 is possible downgrate
  pdfstandard=a-4f,% creates compliance with the PDF/A standard
  uncompress,
 % testphase={block=false},   % <-- required May 2026
}
%
% Refer to documentation (ndsu-thesis-tagged-documentation....) for various options and commands
% Use LuaTeX for compiling (LuaTeX+Biber+LuaTeX+LuaTeX for docs with bibliography)
%
% Created by C. Igathinathane <i.cannayen@nsdu.edu>,  ABEN, NDSU
%
%
%***********************************************
%***********************************************
%******************* START *******************
%\documentclass[ms-thesis,mathdesign,chpterrefs]{ndsu-thesis-tagged}% MS final, 12 pt
\documentclass[ms-thesis,mathdesign,chapterrefs,showgriddm]{ndsu-thesis-tagged}% MS work, chap refs, 12 pt, showgrid darkmode
%\documentclass[ms-thesis,mathdesign,showgriddm]{ndsu-thesis-tagged}% MS work, 12 pt
%\documentclass[showgriddm]{ndsu-thesis-tagged}% PhD final, 12 pt, computer modern
%\documentclass[mathdesign,darkmode]{ndsu-thesis-tagged}% PhD final, 12 pt, computer modern
%\documentclass[mathdesign,chapterrefs,showgriddm]{ndsu-thesis-tagged}% PhD, chap refs, work
%\documentclass[mathdesign,chapterrefs]{ndsu-thesis-tagged}% PhD, final, chap refs, 12 pt

%------------------------------------------------------
\graphicspath{{./figures/}}

%******************* Packages, newcommands, and other customization *******************
\usepackage[style=apa,natbib=true,backend=biber]{biblatex}% works with \citep and \citet commands
% For numeric styles, viz., ieee, numeric, numeric-comp see documentation Sec.

\addbibresource{mybib.bib}% *.bib extension is necessary
%\renewcommand\myspacing{1.9} % 23 lines/page needs 1.9 for thesis

%***** SI units setup and a few custom unit commands *****
\sisetup{group-separator = {,}}% thousands separator comma; space default
\DeclareSIUnit\ft{ft}
\newcommand{\sqft}[1]{\qty{#1}{foot}$^2$\xspace}% have math outside of SI
\newcommand{\cuft}[1]{\qty{#1}{\cubic\ft}\xspace}% SI standard commands

%******************* First and second page material *******************
% Default 2-line title. Otherwise use \setlength{\droptitle}{Xin}; X = -1.57 for 1-line  and -1.47 for 3-line titles
\title{Example NDSU LaTeX Thesis Template for Automatic PDF Tagging in Masters and Doctoral Dissertations}
\author{Samuel Fargo Bison}
\keywords{NDSU, Thesis, Dissertation, Accessible documents, Add your subject-specific keywords}
\date{December 2026}
\progdeptchoice{Department} % Use Department (or) Program
\department{Mathematics}

\cchair{Prof. John Adams} % Use actual committee members' names
\cmembera{Prof. Abraham Lincoln}
\cmemberb{Prof. George Washington}
\cmemberc{Prof. Theodore Roosevelt} % If 3rd not required - delete this line
\approvaldate{12/14/2026}
\approver{Prof. James Garfield}

%******************* Front matter *******************
\abstract{
        {\color{magenta}
        Dissertation template design and Important instructions for using the template:
        \begin{itemize}
        \item The template has been designed to have some initial instructions, usually in \tcb{blue text} or other colored
          text, and followed by filler text in black, just shown as a placeholder
```

```
        \item Look for this \start and populate your text after removing the existing text (instructions and filler text)
        \item Better make a copy of the template and work from there
        \item Read the ``\texttt{ndsu-thesis-tagged}'' class documentation to understand the available commands
          and features, which cannot be covered in this sample template.
        \end{itemize}}
\start\tcb{Students and users may utilize this example as a ``template.'' Make a copy and start working. Most of the codes are
required to write the thesis are employed in this example (*.tex file), and along with the accompanying ``Documentation''
of the ``\texttt{ndsu-thesis-tagged}'' class, all necessary commands, options, and functionalities to generate a
dissertation are covered. As this \LaTeX{} class generates tagged PDF, it is ``future-proof'' and supports the creation of
``Accessible Documents.'' The showgrid option, useful for vertical spacing of items (reference: $\approx$3 grids =
$\approx$0.3 in), can be removed for the final output. Use LuaTeX for compiling (LuaTeX+Biber+LuaTeX+LuaTeX for docs
with bibliography) for better tagging functionality. By the way, the ``\textbf{blue text}'' or other
\textcolor{magenta}{colored} text or \hl{highlights} are used to draw attention!}\\
\indent \tcm{This is the abstract for my thesis. The word limits: 350 for PhD and 150 for MS.} \\
\indent\repText[50]
}%

\acknowledgements{\start I acknowledge people here. \\ \tcb{\emph{Acknowledgements text should be placed here.}} \\
 \indent\repText[100] \\ \indent\repInText[my acknowledgement here][30]
}%

\dedication{\start This thesis is dedicated to my cat, Mr. Fluffles. \\
\tcb{\emph{This section dedicates the disquisition to a few significant people. The text must be double-spaced and aligned
center to the page.} \\ Which is already taken care by this \LaTeX\ class.}
}%

\preface{\start You can put a preface here. \\
\tcb{\emph{This section is optional!}} \\
\indent\repText[120]
}%

\listofabbreviations{ % may use title case; tagged version; usage  \item[X]  -space-  def of X
   \item[\start]  \tcb{delete this list and populate with yours}
   \item[AC]    alternating current
   \item[NDSU]  North Dakota State University
   \item[MC]  moisture content
%------------------------------------------------------
\section{Abstract}
Paper-styled chapters will have abstracts (PDF tag \tcb{H2}). \tcb{This \LaTeX{} \texttt{ndsu-thesis-tagged} class will
automatically tag all elements (H1--H8, paragraph, caption, TH, TD, etc.) of the thesis (when instructions are followed
properly)}. Abstract of this chapter goes here. Have your abstract text here. \repText[10]

%------------------------------------------------------
\section{Section ($\Rightarrow$ 1st level; Title Case; Centered; Boldface) E.g., Science is Key Period}
This is the first section of the thesis (1st level: 1.2. Section, PDF tag \tcb{H2}). \repInText

%------------------------------------------------------
\section{Section E.g., Science is Key Period}
This is the second section of the thesis (1st level: 1.3. Section, PDF tag \tcb{H2}). \repText[18]

%------------------------------------------------------
\subsection{Subsection ($\Rightarrow$ 2nd level; Title Case; Left-justified; Boldface) E.g., Science is Key Period}
This is the subsection text (2nd level: 1.3.1. Subsection, PDF tag \tcb{H3}). \repInText[my `special' input text][11]

%------------------------------------------------------
\mysectop
\subsubsection{Subsubsection ($\Rightarrow$ 3rd level; Sentence case; Left-justified; Boldface; Italics) E.g., Science
is key period}
This is the subsubsection text (3rd level: 1.3.1.1. Subsubsection, PDF tag \tcb{H4}). \repText

%------------------------------------------------------
\paragraph{Paragraph ($\Rightarrow$ 4th level; Sentence case; Left-justified; No bold; Italics)  E.g., Science is key period}
This is the paragraph text (4th level: 1.3.1.1.1. Paragraph, PDF tag \tcb{H6}). \repText[18]

%------------------------------------------------------
\subparagraph{Subparagraph ($\Rightarrow$ 5th level; Sentence case; Left-justified; No bold; Regular) E.g., Science
is key period}
This is the subparagraph text (5th level: 1.3.1.1.1.1. Paragraph, PDF tag \tcb{H7}). \repText[20]

%------------------------------------------------------
\section{Tables and Figures}
This is the third section of the thesis (1st level: 1.4. Section, PDF tag \tcb{H2}). This section illustrates the inclusion of
a simple table (\cref{tab:1}) and a figure shown later.

\voh% shortcut = \vspace{-0.15in}
\begin{table}[ht!]
\centering
\caption{Table captions go at the top of the table. Compatible \texttt{tabular}. This was a long caption of the table included
in the first chapter---so that we see how it breaks into another line and has a single spacing. Usually, tables are of full-width
```

```
and are demonstrated subsequently.}
\label{tab:1}
\pvo
\tagpdfsetup{table/header-rows={1}}% for tagging the header row(s) {1,2}
\begin{tabular}{clr}
\toprule
Number & Month & Days\\
\midrule
\#1 & January    & 31\\
\#2 & February   & 28\\
\#3 & March      & 31\\
\bottomrule
\end{tabular}
\end{table}

\voh
\repText[100]

\tcb{Note: The vertical spacing around (before and after) the float element (figures, tables, schemes, etc.) should be similar
to the regular text with ‘‘double spacing’’ the natural automatic spacing of \LaTeX. This is achieved using the
\texttt{\textbackslash vspace\{\ldots\}} command with $+$ve or $-$ve arguments. Convenient shortcuts were defined.
Observe the source code.}

\tcb{Now the figure (\cref{fig:11}) illustrates an example figure from the \texttt{mwe} package.}

\mytfig{ht!}{0.6}{example-image-duck}{Caption for this example image in this first chapter using the tagged shortcut command
\texttt{mytfig} that has optional vertical spacing before and after the figure caption.}{A yellow duck with a golden crown and
pizza in the belly on a gray background.}{fig:11}% all default options

\vo% shortcut = \vspace{-0.1in}
\repText[58]

%-----------------------------------------------------
\section{Some References as Section or Combined}
Some bibliography commands are parenthetical, like these  \citep{texbook,lcompanion,latex2e,knuth1984,cannayen2011latex,
kopka2004guide}. The authors \citet{texbook,lcompanion,latex2e,knuth1984,amsthm2017,calvo2004using,cannayen2011latex}
have something to do with \LaTeX. For most bibliography citations and list creation, these two commands are sufficient.

%-----------------------------------------------------
\checkEndRefsection%%% Don’t delete this! Command given at the end of each chapter
%-----------------------------------------------------
```

Note: Codes of the chapters 2 and 3 + Appendix B are not shown. Download the “`ndsu-example-tagged.tex`” source code and other resources to start working (*.tex + *.cls + *.bib + figures folder = zip file available)

```
%---------------------------------------------------------------------------
%****************** Bibliography handling ******************
% Combined unnumbered Reference  chapter - automatically generated
\checkMakeCombinedReferences%%% Don’t delete this - can be moved last
%---------------------------------------------------------------------------

%*******************************************************************
%********************** Named appendix A **********************
%*******************************************************************
\namedappendices{A}{Named first appendix}

\checkBeginRefsection%%% Don’t delete this! Command given at the start of each chapter

\noindent\start \tcb{Start writing your appendix following the template and examples given here.}

\tcb{Appendix material can be included here. First, a paragraph of text and then an example figure (fig.~\ref{fig:a1}).}
\hl{If bibliography citations are required, then \cmd{checkBeginRefsection} and \cmd{checkEndRefsection} blocks can be added to
each appendix as coded in the regular chapters (as included here). When coded, it will create the ‘‘Reference’’ section listings.}

%-----------------------------------------------------
\section{Appendix A - Section With Figure}
\kant[9]

\mytfigap{h!}{0.45}{example-image-golden}{A golden ratio rectangle image from the \texttt{mwe} package.}{A gray rectangle drawn
with the golden ratio.}{fig:a1}% all default options

\repText[58]

%-----------------------------------------------------
\mysectop
\section{Appendix A - Section With Non-floating xltabular Table}
\tcb{A complete full-width table utilizing the tagging compatible ‘‘xltabular’’ package is shown below (table.~\ref{tab:a1}). It is
also known that appendix tables are exclusive and should be coded using the ‘‘\texttt{appendixtable}’’ environment (as shown below)}.
```

```
\vo
\begin{appendixtable}[h]
\caption{A complete full-width xltabular example --- Courtesy MMS.} \label{tab:a1}

\captionsetup[table]{margin={1.5pt, 0cm}}% check if required
\renewcommand{\arraystretch}{0.5}
\vh
\tagpdfsetup{table/header-rows={1}}% for tagging the header row(s) {1,2}
\begin{xltabular}{\textwidth}{
>{\hsize=0.5\hsize\RaggedRight}X           |
>{\hsize=1.2\hsize\Centering}X                |
>{\hsize=1.3\hsize\RaggedLeft}X}

\toprule
First - left & Second - centered & Third {X} column right aligned\\
\midrule
a & b & c\\
a & b & c\\
\midrule
a & b\\
a & b & c\\
\cmidrule(lr){2-3}
& \multicolumn{2}{c}{The combined block of 2nd and 3rd columns}\\
\cmidrule(lr){2-3}
a & b & c\\
d & e & f\\
\bottomrule
\multicolumn{3}{@{} p{\textwidth} @{}}{\vspace{0in}
Note: In many cases, the word ``Note'' is not required. Again, the vertical lines are simply given for illustration purposes,
as in professional tables, they are not used.}
\end{xltabular}
\end{appendixtable}

\vrh \tcb{For other types of tables and figures, such as landscape tables, long tables, landscape long tables, landscape figures,
subfigures, subfigures spanning multiple pages, and multiple figures in landscape, see NDSU-Thesis-Extended code and output.}

%------------------------------------------------------
\subsection{Appendix A - Subsubsection}
\repText[218]

%------------------------------------------------------
\section{Appendix A - Subsection With Listing}
\tcb{See Listing~\ref{lis:21} preamble for information.} \repText[68]

% Step 1 of 2:    Code the listing appendix caption [before vspace] {caption} [after vspace]
\mytliscapap{Java code listing for finding all prime numbers from 1 to 100.}{lisa:1}

% Step 2 of 2:    Code the listing using a simpler mytlisting environment
% Other options [font sizes] [frame=none, single] [numbers=none, left, right] [xleftmargin]
\begin{mytlisting}[fontsize=\normalsize, frame=single, numbers=none, xleftmargin=0pt]
public class PrimeFinder {
    public static void main(String[] args) {
        int limit = 100;
        System.out.println("Prime numbers from 1 to " + limit + ":");
        for (int i = 1; i <= limit; i++) {
            if (isPrime(i)) {
                System.out.print(i + " ");
            }
        }
    }
    public static boolean isPrime(int n) {
        if (n <= 1) return false;
        for (int i = 2; i <= Math.sqrt(n); i++) {
            if (n % i == 0) return false;
        }
        return true;
    }
}
\end{mytlisting}

\repText[38]

%------------------------------------------------------
\checkEndRefsection%%% Don't delete this! Command given at the end of each chapter
%------------------------------------------------------
```

Note: Other codes not shown

```
\closeappendices  % Refer to documentation Table 3 for proper closing

\end{document}

%******************* END *******************
%*********************************************
```

The example thesis template code (`ndsu-example-tagged.tex`), when downloaded with resources compiled, will produce the output shown in Figure 1. The example source code serves as a lean template with meaningful and dummy text. The citations in the text and reference listing were automatically generated according to the selected reference style.

A lightweight source file named "`ndsu-barebone-tagged.tex`," which can be used to try out things conveniently in the actual NDSU thesis environment, was also included in the package folder. Things tested here (including the bibliography) can be readily inserted into the original thesis/dissertation document.

In addition, another extended file named "`NDSU-Thesis-Extended.tex`" (link:) containing several additional comments (listing the various options of the class) and features was made available as a supporting document. Most of the requirements of the students for disquisition will be covered with these two example source code files.

8 PDF Tagging DocumentMetadata

The following `\DocumentMetadata` code should be the first statement before the usual `\documentclass{...}` command. This command contains the PDF tagging instructions that will convert a regular LaTeX document (untagged PDF) into a tagged PDF (accessible) document—provided the commands used are from the tagging-compatible packages (as discussed earlier).

```
\DocumentMetadata{
 tagging=on,
 lang=en-US,
 tagging-setup={math/setup=mathml-SE,math/alt/use},
 pdfversion=2.0,
 pdfstandard=ua-2,
 pdfstandard=a-4f,
 uncompress
}
```

9 Documentclass Options

These are the options passed to the documentclass command while calling the class. These options essentially affect the whole document, and a default behavior (no options specified, shown below) was also valid.

`\documentclass{ndsu-thesis-tagged}`

The default behavior with no `[options]` specified, as shown above, produces a Ph.D. dissertation in 12 pt font size with auto-numbered headings and justified text in Computer Modern font. However, the command:

`\documentclass[ms-thesis,12pt,nonumber,nojustify,draft,showframe,mathdesign]{ndsu-thesis-tagged}`

produces an M.S. thesis in 12 pt font size, with unjustified paragraphs, text with unnumbered headings in draft mode, in Bitstream Charter font, and shows the frame using the set margins. The order in which these options are passed does not matter.

9.1 Disquisition degree and type

One of the important options of the class is the degree type. By default, this class assumes the document is a Ph.D. dissertation. The other types of available degrees and disquisition types are given in Table 1.

Table 1: Options for degree and disquisition types

Option	Degree	Disquisition type
[phd]	Ph.D.	Dissertation (default)
[ms-thesis]	M.S.	Thesis
[ms-paper]	M.S.	Paper
[ma-thesis]	M.A.	Thesis
[ma-paper]	M.A.	Paper

9.2 Collection of all options of documentclass

The various documentclass options, which are related to font size, section numbering, text justification, showing helpful frames and grids, font styles, degrees and types, and bibliography handling, as well as the defaults (shown in blue), are listed in Table 2. If any other undefined options or mistyped options are ignored, the defaults will be used for the output. Details of the options are described subsequently.

Table 2: List of all documentclass options (fonts and others) and the defaults already loaded in NTTC

10pt	11pt	12pt (d)	nonumber	numbered (d)
nojustify	draft	showframe	showgrid	computermodern (d)
bookman	charis	charter	clearsans	cmbright
firasans	helvet	kpfonts	kurier	libertine
lxfonts	mathdesign	mathptmx	mlmodern	newcent
newcomputermodern	newpx	newtx	opensans	palatino
sansmathfonts	times	tgbonum	tgpagella	tgschola
utopia	chapterrefs	phd (d)	ms-thesis	ms-paper
ma-thesis	ma-paper	darkmode	showgriddm	

Option (d) - default options already loaded (need not specify them in the documentclass)

9.3 General requirement for thesis—formats, margins, and spacings

The most general and mandatory requirement for the thesis is adherence to the rules set by the school. These include the (i) formats (headings, captions, general test, and appropriate cases and style of headings and text), and (ii) margins and spacings (margins of page [1 inch all around]; double-spacing for general text, and single spacing for captions and table contents; paragraph indent [1/2 inch]; spacing around all elements [tables, figures, schemes, listings] should be the same as the double-spacing followed with the general text).

The NTTC is developed to take care of these requirements; however, as LaTeX algorithm typesets elements automatically based on the available space and introduces additional space (as seen below the Table 2 and the heading of this section). This additional space can be corrected using \vspace{} command with −ve values to suit the requirement. The document "ndsu-thesis-vertical-spacing" (link:) added in the package exclusively addresses this issue. It will also be useful to utilize the documentclass `showgrid` and `showframe` options (fig. 2) to visualize the margin for page elements. The `showframe` option will be helpful, specifically for the landscape page elements, for correct resizing so that everything fits as it should.

When there is no room to fit an element (table or figure) in the given space, it will be floated to the next page, leaving a blank space on the current page; such blank spaces are "okay" to have, as these are unavoidable. Sometimes, it is possible to move the text after the float element and fill this blank space, as long as the content allows such a move (e.g., major headings or other floats cannot be moved to fill this space).

10 Preamble Information

10.1 General information - packages and shortcuts

If your disquisition requires the use of additional LaTeX packages, macro files, or other commands, include them in the preamble. Packages such as `natbib` (author-year and number style citations) or other competent reference handling packages and their options can be loaded. Similarly, mathematical theorem environment-related commands (theorem, corollary, lemma) through `\newtheorem` of `amsthm` package, `caption` package setups through `\captionsetup[type]{options}`, where `type` = table or figure, and other shortcuts for repetitive longer commands or text can be defined in the preamble. As these are specific to the users and the requirements vary with the users of different specializations, these were not included in the class. Therefore, suitable packages, commands, and shortcuts can be defined by the users.

10.2 Dissertation front pages

Before issuing the `\begin{document}` command, several pieces of dissertation prefatory (preamble) information are available.

10.2.1 Title

Include the title of the disquisition using the `\title{...}` command. This is required. The title is optimized for a title text that is 2 lines long. For others, the `\vspace{...}` command with +ve or -ve values may be issued as the `\title` argument.

10.2.2 Author

Include the full name of the disquisition author using the `\author{...}` command. This is required.

10.2.3 Keywords

The keywords are required and are generated using the `\keywords{...}` command. The keywords can be general (e.g., NDSU, Doctoral dissertation) as well as specific to the subject of the dissertation. The keywords will not be displayed in the dissertation output, but will be shown in the PDF Document Properties, along with the title, author, and language (supplied by NTTC). This is necessary for accessibility checking.

10.2.4 Major department/program choice

Specify whether it is "Department" or "Program" that is applicable using the `\progdeptchoice{...}{...}` command. This will ultimately produce "Major Department:" or "Major Program:" based on the input choice. This is required.

10.2.5 Department or program

Include the name of the major department or program using the `\department{...}` command. This is required.

10.2.6 Degree option

If the major department or program has a degree option, indicate this using the `\degreeoption{...}` command. This is optional.

10.2.7 Date

Include the date of the final examination using the `\date{...}` command. The accepted format of this date is *month year* as: `\date{December 2026}`. This is required.

10.2.8 Examining committee

Include the Chair (or Co-Chairs) and members of the examining committee using separate commands. The `\cchair{...}` command is used to indicate the committee Chair. Use `\cochairZ{...}` to indicate any committee Co-Chair members, where `Z` is `a` or `b`. This class does not support more than two Co-Chairs and four Committee Members. Use the `\cmemberX{...}` to indicate other committee members, where `X` is `a`, `b`, `c`, or `d`. Use only as many of these commands as needed to list all committee members.

10.2.9 Approval information

Use the `\approvaldate{...}` command to include the full date of disquisition approval (i.e., month/date/year). This date is generally the date the thesis was approved by the Department Chair following the defense, after the approval (usually electronically) of all committee members. Use `\approver{...}` to include the Department Chair who approved the disquisition. Both commands are required.

10.3 Dissertation front matter

10.3.1 Abstract

Use the `\abstract{...}` command to include the disquisition abstract. Abstracts for doctoral dissertations must use 350 words or less. Abstracts for master's papers or master's theses must use 150 words or less. This is required.

10.3.2 Acknowledgements

If the disquisition includes acknowledgements, include them using the `\acknowledgements{...}` command. This is optional.

10.3.3 Dedication

If the disquisition includes a dedication, include it using the `\dedication{...}` command. This is optional.

10.3.4 Preface

If the disquisition includes a preface, include it using the `\preface{...}` command. This is optional. The NDSU guidelines state:

> "*The Preface can provide an autobiographical account of how the disquisition came to be, or include a significant quote that drove your research. Follow the General Requirements for font, spacing, and page numbers for prefatory materials.*"

11 Automatic Components

Several automatic components will be generated, as a part of the front matter, based on the source code of the dissertation, and are briefly described. Based on the department, requirement, and style of the thesis, some of the items, such as lists of abbreviations, symbols, and appendix tables and figures (Secs. 11.3–11.5) may be dropped from the coding.

11.1 Table of contents

The table of contents (TOC) gets automatically generated with entries up to three levels of sections (`\myheading{...}` or `\mypaperheading{...}`, `\section{...}`, and `\subsection{...}`). The dissertation may have further levels of sections, but they are not shown in the TOC.

11.2 List of tables, figures, schemes, and listings

The list of tables (LOT), list of figures (LOF), list of schemes (LOSH), and list of listings (LOL) will be generated based on the table, figure, and scheme full captions in the `table`, `figure`, and `scheme` environments. New commands for handling figures such as `\mytfig{...}`, `\mytfigls{...}`, `\mytsch{...}`, and `\mytschls{...}` with their own `[optional]` and `{mandatory}` arguments to adjust the position of the float and caption with respect to figure and scheme elements were defined. The letter '`t`' in the command name refers to `tagging` that is followed in NTTC. More and similar commands are available (`\mytfigap{...}`, `\mytschap{...}`, etc.) for the appendix as well.

11.3 List of abbreviations

The collection of abbreviations used in the dissertation can be made into a list of abbreviations (LOA) using the `\listofabbreviations{...}` command. The LOA was coded as a tagging-compatible "`list`" definition environment, and the `\item[...]  definition` coding format should be used. The final collection should be alphabetized. Lowercase may be used for the definitions unless items are proper names. A five-entry example of the LOA code and the output is shown below:

```
\listofabbreviations{
\item[AC] alternating current
\item[NDSU] North Dakota State University
\item[MC] moisture content
\item[US] United States
\item[ZL] zeta level
}
```

LIST OF ABBREVIATIONS

AC .... alternating current
NDSU .... North Dakota State University
MC.............................. moisture content
US .... United States
ZL............................... zeta level

11.4 List of symbols

The collection of all technical symbols used in the dissertation, usually coded in "math" mode, can be made into a list of symbols (LOS) using the `\listofsymbols{...}` command. This collection should be alphabetized, and math mode should be used as required. Again, lowercase may be used for definitions unless items are proper names. Again, this is a tagging-compatible list. A five-entry example of LOS also using the `siunitx` package, and the output is shown below:

```
\listofsymbols{
\item[$A$] area (\unit{\m\squared})
\item[$c$] speed of light (\qty{299792458}{\m\per\s})
\item[$e$] Euler's constant (\num{2.718281828})
\item[$r$] correlation coefficient (dimensionless)
\item[$R^2$] coefficient of determination (dimensionless)
\item[$t$] time (\unit{\s})
}
```

LIST OF SYMBOLS

A .... area (m^2)
c .... speed of light (299,792,458 $m\,s^{-1}$)
e .... Euler's constant (2.718,281,828)
r .... correlation coefficient (dimensionless)
R^2 .... coefficient of determination (dimensionless)

11.5 List of appendix tables, figures, schemes, and listings

The list of appendix tables (LOAT), list of appendix figures (LOAF), list of appendix schemes (LOAS), and list of appendix listings (LOAL) will be generated based on their respective environments with full captions. Shortcut commands, similar to the regular chapters (e.g., `\mytfigap` and `\mytfiglsap` were developed with caption placement.

The `\closeappendices` command should be issued at the end of the last appendix, which ensures the automatic creation of the LOAT, LOAF, LOS, and LOL when the appendices have tables and figures. If the appendices had only tables or only figures, then the commands `\closeappendixtables` or `\closeappendixfigures` should be used to avoid blank entries, and if no tables or figures are used in appendices, then these commands should not be used or commented out, and other details are shown in Table 3.

12 Basic Components and Commands

12.1 Beginning the document

After including the necessary preamble information, use `\begin{document}` to start the document. This command automatically generates the necessary cover pages and other automatic components. The usual `\maketitle` command should not be used, as it was already issued in the class.

12.2 Headings

Major headings (e.g., chapters) are issued using the `\myheading{...}` command. This command supersedes the usual `\chapter` command, which should not be used. The following shows the hierarchy of headings:

`\myheading{...}`	Produces all-caps chapter headings automatically - 0th level
`\mypaperheading{...}{...}`	Produces all-caps paper chapter headings with footnote - 0th level
`\section{...}`	Produces centered, **bold** headings (use title case) - 1st level
`\subsection{...}`	Produces left-aligned, **bold** headings (use title case) - 2nd level
`\subsubsection{...}`	Produces left-aligned, **bold**, *italic* headings (use sentence case) - 3rd level
`\paragraph{...}`	Produces left-aligned, *italic* headings (use sentence case) - 4th level
`\subparagraph{...}`	Produces left-aligned, regular headings (use sentence case) - 5th level

Each `\myheading{...}` or `\mypaperheading{2 args}` command starts a new page and entry in the TOC. The regular chapter command is simple and takes one argument, which is the title of the chapter as `\myheading{title}`. The paper-styled chapter takes three arguments that address the title and footnote as `\mypaperheading{title}{footnotetext}`. The paper-style chapters have footnotes, and the title has a numbered footnote mark. The `title` is common to both styles and will be rendered as all-caps irrespective of the input. The automatically generated footnote mark, an integer sequentially incremented, is associated with the footnote text, which is displayed at the end of the page. The footnote text usually details the publication information with the roles of authors (refer to Grad School information for further details).

Based on the latest instruction, only the "previously published papers" should have the footnotes, which detail the citation of the paper and the roles of the authors, and other "papers in preparation/submission" need not have the footnotes. So for (i) chapters featuring "published papers" use `\mypaperheading{...title ...} {footnotetext of publised paper + citation + authors contribution}`; and (ii) for "unpublished papers" use simply the chapter style `\myheading{...title...}`.

In general, the paper-styled chapter requires an "Abstract" section, while the regular chapter (e.g., Introduction, Review of Literature, etc.) does not. The class is coded to produce a consistent space between the title and the text (or section) below the title; however, when necessary `\vspace{+ve or -ve}` can be issued before the plain introductory text or section command to adjust this vertical space. The `\showgrid` documentclass option, if required, can be used to judge the vertical spacing.

Instances of `\subsubsection{...}` and higher levels do not appear in the TOC, though they are included in the document. Other than the chapter headings, the rest of the item headings should be coded by the user manually with appropriate capitalization (title and sentence cases).

12.3 Dummy text and images

Users will be curious to see what their thesis/dissertation will look like quickly, without using the actual texts and figures. The class comes loaded with necessary packages such as `kantlipsum` (for dummy text—philosophical prose paragraphs in English) and `mwe` ("minimal working examples" for dummy images).

These will help visualize the whole document (fonts, spacing, and layout) with minimal effort, and this is a common practice among typesetters to use such dummy text and images.

Commands like `\kant[1]` or `\kant[4-8]` will produce single or multiple dummy text paragraphs. Another method in NTTC is (1) `\repText`[...], which takes a number as an optional argument and generates the word "text" that many times; and (2) `\repInText`[...][...], which takes two optional arguments (any input text) and a number and generates the input that many times. These commands were used in the example thesis.

Similarly, dummy images included in the (1) `mwe` package or (2) through the original "tex" distribution (link: Stackexchange-630673) using the `grep` code, the list of several existing intersting images can be generated and can be accessed using their specific names and can be used as the image argument in the `\includegraphics` command, which means that the user need not use their images. Some of the commonly used examples images from `mwe` are: `example-image`, `example-image-a`, `example-image-b`, `example-image-c`, `example-image-16x10`, `example-image-golden`, `example-image-pl ain`, `example-image-duck`, and `example-image-empty`. Refer to the documentation of the package for further information.

Similarly, some of the images from the LaTeX distribution are: `CleanEasy-logo1.png, CleanEasy-logo4.png, NPBT_SC_logo.png, NPBT_FOM_ifes_logo.png, ciad-title-background-left1.png, ciad-title-background-left4.png, ciad-title-background-left5.png, ciad-title-background-left8.png, blueshade.png, crinklepaper.png, pink_marble.png, ghsystem_flame.png, ghsystem_skull.png, ep-ball-01.png, ep-ball-05.png, ijsra_logo.png, utbmciadreport-ciad-frontbackground.png, context-version.png, mill.png, stex-dangerous-bend.png, sTeX-logo.png, op-ring.png, op-slides-bg.png, fancymag-logo.png, fancymag-m_sig.png, fancymag-msg.png, fancymag-msg1.png, CaJ-TaroTv1-1AT.png, CaJ-TaroTv1-21AT.png, CaJ-TaroTv1-2C.png, CaJ-TaroTv1-10C.png, CaJ-TaroTv1-VC.png, CaJ-TaroTv1-VK.png, and CaJ-TaroTv1-Dos.png`

12.4 Tables

Different kinds of tables, such as simple table without caption (`tabular`), table with caption (`table`), table spanning entire text width (`xltabular`), table spanning multiple pages (`longtable, xltabular`), and table in landscape page (`pdflscape`) can be coded following the documentation of respective packages, and no shortcuts were defined as they were not practical. Using `booktabs` package, the professional quality tables (section 9.1) can be created. Examples of using these commands can be found in the example and/or extended example theses (link:) of the class. The guiding principle is to have a compact table based on the number of columns and avoid excessive space between columns, as well as font sizes not too small when there are more columns of information. The command:

```
\tagpdfsetup{table/header-rows={1}}% for tagging the header row(s) {1,2}
```

should be included before the `tablular` or `xltabular` environments to tag the table header rows (TH), while the other rows will be tagged as the table data (TD).

12.4.1 Tables with fewer columns

Compact tables with fewer columns are common and readily made by the common `tabular` environment. Where the columns will be based on the width of the widest entries, and the columns will be naturally spaced and result in a compact table with the total width usually less than the textwidth. No special action is necessary to make these tables. Tables of fewer columns and narrower widths need to be positioned on the page consistently. Either all of them are left-justified or centered. Footnotes corresponding to the width of the table can be coded through the `\multicolumn{no of cols}{lcr}{text}`\\for single line items or `\multicolumn{no of cols}{p{dimension}}{text}` for footnotes that run like a paragraph. The width of the footnote is controlled by the amount of text or the dimension of the paragraph (refer to the "`NDSU-Thesis-Extended.tex`" (link:) for example codes).

12.4.2 Full-width tables—`xltabular` environment from `xltabular` package

When more columns are present in the table and can fit in the textwidth, it is better to code as automatic full-width tables. A quick and efficient method of creating tables that automatically span the entire textwidth is the use of `xltabular` environment instead of the usual `tabular` inside the `table` environment. This `xltabular` environment uses a special column justification code `X` (default & options). If one `X` column, along with other usual `l, c, r` columns, were issued, the table will automatically become full-width.

If all columns are of type `X`, then the total width specifications of all columns should be equal to the total number of columns. The following formula can be used to find the `\hsize` actual value (decimal value calculated) of the X column from the proportional size of the column.

$$\text{Actual column size:} \quad x_i = P_i \times \frac{n_c}{\sum_1^{n_c} P_i} \tag{1}$$

where, x_i is the actual size of the column (e.g., `\hsize`=0.56`\hsize`), P_i is the proportional width desired (e.g, 2, 3, 1.5), and n_c is the number of columns. It can be observed that it is intuitive to select the n_c and P_i, but the formula will calculate the actual width (x_i). See the example thesis where X column size is specified using x_i and the justification in the form of new columns definition (e.g., Y, Z based on X).

The footnote is coded using multicolumn cells (see the example and refer to the documentation). The `xltabular` is a non-floating table environment that can produce both full-width and long tables.

12.5 Figures and schemes

The **figures** usually are pictures, photographs, drawings, maps, illustrations of samples, fields, instruments, structures, methods; graphs or plots of measurements, results; or anything graphically depicted to convey the thoughts. However, **schemes** are systematic plans for implementing an idea or concept, usually used to depict a process flow and the steps involved, and often involve "arrows" connecting one step to the next. Both figures and schemes should have "Alternate Text" (alt-text), explaining the content of the graphics to visually challenged individuals. Examples of schemes are chemical process diagrams, sets of chemical reaction pathways, flowcharts (process and computer algorithms), electrical circuits, block diagrams connected by arrows, and so on.

It should be noted that the long format coding of figures using "`figure`" environment with `\includegraphics{...}` centering, resize, caption and labels is the direct approach and is always available to the users. Similarly, the schemes are coded using "`scheme`" environment. By default, the schemes are labeled as **Schematic** in their caption.

12.5.1 Shortcuts for figures and schemes—with direct and optional arguments

For convenience, a set of shortcuts (below), with 9 arguments (LaTeX3 style arguments: O m m m O m O m m), is defined for the accessible documents with figures (tagged PDF).

```
\mytfig[0.09in]{ht}{0.7}{image1o.jpg}[0.025in]{Figure caption with placement option.}[-0.08in]{Alt-text.}{fig:1}
\mytfigls[0.09in]{p}{0.7}{image1o.jpg}[0.025in]{Figure caption with placement option.}[-0.08in]{Alt-text.}{fig:2}
\mytsch[0.09in]{ht}{0.7}{image1o.jpg}[0.025in]{Scheme caption with placement option.}[-0.08in]{Alt-text.}{sch:11}
\mytschls[0.09in]{p}{0.7}{image1o.jpg}[0.025in]{Scheme caption with placement option.}[-0.08in]{Alt-text.}{sch:12}
```

where: `t` - tagged, `fig` - figure, `ls` - landscape, and `sch` - scheme.

These commands specify (1) [optional] vertical spacing above the actual float (moving it up or down with respect to the text before the float), (2) placement specifiers `[h, t, p, b, !]` but not `H`, which is incompatible, (3) size factor for the image, (4) input file, (5) [optional] vertical spacing below the float and above the caption (moving the caption up or down with respect to the float above), (6) caption, (7) [optional] vertical spacing below the caption and above the following text (moving the following up or down with respect to the caption above), (8) alt-text, and (9) label. The command logic and style are similar for regular and landscape figures and schemes. The commands will also work without the three optional arguments, in which case they use the specified default values (below) of the optional arguments. Users, as needed, include any of these unambiguously positioned optional arguments with specific vspace values with a goal to achieve a 3 grid (0.3 in) spacing above and below the float with surrounding text and about 1 or 1.5 grid (1–1.5 in) between the float and the caption. These commands automatically generate TOC entries, a respective lists of entries, and bookmarks.

12.6 Listings

Listings are an integral part of the dissertation, and help in enhancing the reproducibility. Listings can be coded in the chapters and in appendices. Coding listings is a two-step process, namely, caption and actual listing (see example thesis and documentation). The captions that got listed (LOL, LOAP) are separately coded (each for the body text and the appendix) using the following commands. For now, as the caption can be read by the screenreader, no "alt-text" is coded with the caption.

```
\mytliscap[-0.1in]{Caption for the listing.}[-0.275in]{lis:1}
\mytliscapap[-0.1in]{Caption for the appendix listing.}[-0.275in]{lis:a1}
```

where: `t` - tagged, `lis` - listing, `cap` - caption, and `ap` - appendix.

Because of the tagging incompatibility of the versatile `listing` package, the listings are coded using the `fancyverb` package. A new `mytlisting` environment was developed to produce the source code. The other options, which will produce a customized listing for practical purposes, available are: [font sizes] [frame=none, single] [numbers=none, left, right] and [xleftmargin=size]. See the example thesis output for this 2-Step process.

```
\begin{mytlisting}[fontsize=\normalsize, frame=single, numbers=left, xleftmargin=0pt]
    ... your listing code ...
\end{mytlisting}
```

12.7 Captions

Because of the way spacing is handled, captions in `table` environments must appear at the top of the table, while captions in `figure` environments must appear at the bottom of the figure. NDSU style follows left-justification and bold font (only for the labels) for both captions. If you use both `\caption` and `\label` commands in these environments, the `\caption` command must come before the `\label` command to ensure the environment is numbered correctly. The captions are coded in such a way that shorter ones are centered and longer ones are left-justified; however, by default, the table captions are left-justified and can be changed, as outlined subsequently, to fit the requirement. The style of the caption can be basic or specific to the department, usually following the parent technical society's leading journal or style guide. The style of labeling (e.g., regular *vs* bold *vs* italic, naming: Fig. *vs* fig. *vs* Figure, etc.) can also be adopted from the leading journal. The various options available for caption using the `caption` package (already loaded) can be set through the `\captionsetup{...}` command. The common options are `position, skip, belowskip, aboveskip, font, labelfont, labelsep, singlelinecheck, format, justification,` and so on.

12.8 Equations

About handling equations, the NDSU's guidelines state "When coding equations, the guidelines call for the equation to be center-aligned, with the equation number aligned flush with the right margin." The strong suit of LaTeX is the professional manner it typesets the equations and mathematical elements. Show below is the distance formula that was defined and referred [eq. (2)], which satisfies NDSU's guidelines:

$$\text{Distance formula:} \quad d = \sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2} \tag{2}$$

where, d is the distance; and x_1, y_1, x_2 and y_2 are the coordinates of the two points. The equations can be displayed [e.g., eq. (2)] produced by $$...$$ or \[...\] or `equation` environment; and the inline as: $ax^2 + bx + c = 0$ produced by $...$ or \(...\). There exist several other commands available to produce equations through several packages (e.g., `align, array, eqnarray, gather, split, multline`), and any imaginable mathematical information can be coded. Also, LaTeX supports a huge list of symbols (Refer link: The Comprehensive LaTeX Symbol List, which contains 25,731 symbols with 602 pages) that can be used in general text or equations.

12.8.1 Shortcuts for equations—properly spaced vertically

LaTeX engine sets an extra little vertical spacing around equations and non-textual elements (e.g., tables, figures) to make them stand out from the regular text, which is an expected and normal behavior. However, the NDSU guidelines require no additional vertical spacing (same double line spacing) around equations. This additional spacing can be manually corrected by using a $-$ve valued argument in the `\vspace{...}` command before and after the elements as required (section 13.7 for details).

For convenient and automatic correct vertical spacing around equations, the following direct and starred versions of shortcut commands were coded for use in the class (font 12 pt, mathdesign): `\myteqn{ }`, `\myteqn*{ }`, `\mytfraceqn{ }`, `\mytfraceqn*{ }`, `\mytalign{ }`, `\mytalign*{ }`, `\mytgather{ }`, `\mytgather*{ }`, `\mytsplit{ }`, `\mytsplit*{ }`, `\mytmultline{ }`, and `\mytmultline*{ }`. The arguments in these commands are the actual codes of the equation(s) without their environment (already included) in these shortcuts (shown below). Label commands can be included in unstarred shortcuts as usual.

Code

```
\myteqn{
E = m \times c^2
\label{eq11}
}
```

Output

$$E = m \times c^2 \tag{1.1}$$

Code

```
\mytalign*{
E_i &= m_i \times c^2 \\
y &= Ax + B
}
```

Output

$$\begin{aligned} E_i &= m_i \times c^2 \\ y &= Ax + B \end{aligned}$$

As is known, the direct versions of the shortcuts will produce the equation number while the starred (*) versions will not. A similar coding applies to all the aforementioned defined shortcuts, and the produced output will fit well with proper vertical spacing with text around them. For other specialized equation environments, other requirements, or when things do not fit well, the manual method of issuing the appropriate $-$ve or $+$ve valued arguments with `\vspace{...}` command can be followed.

12.9 References/bibliography

The two most common bibliography management systems (BMS) are BibLaTeX and BibTeX; the former being modern and highly versatile, and the latter being simpler. BibLaTeX is recommended as the BMS of choice because of its direct usage, versatility, and future-proof capabilities. Reference or bibliography chapter or section can be combined into a stand-alone chapter (whole), or the reference listing can be included in all individual chapters. The bibliography listing in individual chapters, sometimes desired by the user, can be easily coded using the advanced BibLaTeX, which is also coded in the class. Both systems (individual or whole) use the same reference data in the form of `*.bib` file. The authors recommend using any of the systems by appropriately employing the system-specific commands. Various details of handling bibliography are presented in Section 13.8, and for bibliography compilation issues see Section 13.11.

12.9.1 Cite while you write (CWYW) using natbib

The `natbib` package for bibliography management is widely used and very stable, and follows the CWYW paradigm. The package produces both author-year and numerical citations. The commands like `\citep{...}` citation in parentheses and `\citet{...}` citation in running text are quite useful in particular. These commands will produce the following outputs, for example: "(Author et al., 2022)" and "Author et al. (2022) found ...". More use of `\citep{...}` leads to a "tigher" compact style of writing, which is often preferred. Different reference listing styles can be loaded both for BibLaTeX and BibTeX BMS.

12.10 Appendix

12.10.1 Single and multiple named appendices

If the disquisition includes a single appendix or multiple named appendices, one of two commands must be used to produce them. If the dissertation has only one appendix, use the `\appendix` command to begin it. This command generates an unlettered APPENDIX chapter that can have sections, subsections, and so on, as well as tables, figures, and other elements.

If multiple named appendices are necessary, use the `\namedappendices{...}{...}` that can also contain other elements. The following are two examples of the named appendices:

```
\namedappendices}{A}{First named appendix title here}
\namedappendices}{B}{Second named appendix title here}
```

These appendix commands are optional, but are required if the disquisition includes an appendix. The appendix must follow the unnumbered REFERENCES chapter. NDSU's guidelines on appendices only allow named appendices with letters (e.g., "APPENDIX A", "APPENDIX B"), while numerical or other styles ("APPENDIX 1", "APPENDIX 2", "APPENDIX I", "APPENDIX II", and so forth) are not accepted.

It is necessary to generate the listing of appendices in the TOC up to the subsection level (A.1.1), similar to the regular chapters. To achieve this, the necessary codes were included in the class.

12.10.2 Appendix basic figures and tables

If the appendix contains figures or tables, use the `appendixfigure` and `appendixtable` environments to generate them. These special environments have separate counters for appendices and generate LOAT and LOAF entries correctly after the TOC. The usual `figure` and `table` environments in appendices will not add the entries in LOAT and LOAF as expected (will only go to LOT and LOF). The same rules for centering, captions, and labels used in normal `figure` and `table` environments apply to `appendixfigure` and `appendixtable` environments.

As the coding for figures is simple, shortcuts for `appendixfigure` are possible with basic commands, while shortcuts for `appendixtable` are not and need to be done in the usual manner. Similar to figures and schemes handled in the regular chapters (section 12.5), single command shortcuts for appendix figures and appendix landscape figures were defined. These commands follow the exact logic and pattern (9 arguments with 3 optional). Following are the examples of figure shortcuts for appendix regular and landscape figures:

```
\mytfigap[0.09in]{ht}{0.7}{image1o.jpg}[0.025in]{Figure caption with placement option.}[-0.08in]{Alt-text.}{fig:a1}
\mytfiglsap[0.09in]{p}{0.7}{image1o.jpg}[0.025in]{Figure caption with placement option.}[-0.08in]{Alt-text.}{fig:a2}
\mytschap[0.09in]{ht}{0.7}{image1o.jpg}[0.025in]{Scheme caption with placement option.}[-0.08in]{Alt-text.}{sch:a11}
\mytschlsap[0.09in]{p}{0.7}{image1o.jpg}[0.025in]{Scheme caption with placement option.}[-0.08in]{Alt-text.}{sch:a12}
```

where: `ap` - appendix.

12.10.3 Appendix advanced elements—longtables, listings, schemes, and multipage figures

Handling of advanced elements in appendices (e.g., longtables, code listings, schemes, single-page subfigures, and multipage subfigures) needs additional consideration as these are handled only through `appendixfigure` and `appendixtable` environments—as of now in the class. This means listings, schemes, and subfigure types will be considered as entries of `appendixfigure` and longtables as `appendixtable`, and these elements will be listed in the LOAF and LOAT only, respectively. As separate entries will make it bloated, the above strategy will be a compact solution.

12.10.4 Closing appendices and creating TOC, LOAT, and LOAF

In the class, the creation of LOAT and LOAF as well as their TOC entries requires special consideration. When the last appendix has at least one table and one figure, the TOC, LOAT, and LOAF will be automatically generated without intervention. However, when the last appendix does not have at least a table, or a figure, or both (even though the previous appendices had them), the corresponding LOAT or LOAF and relevant TOC entries will not appear. The solution for the automatic creation of these items, irrespective of the contents of the last appendix is issuing the relevant commands such as `\closeappendixtables`, `\closeappendixfigures`, or `\closeappendices` in the overall code as the last command before `\end{document}` and after the last appendix code. Several scenarios occur due to the presence or absence of appendix tables and figures in the last or previous appendices, and the last command (or) none to be used are presented in Table 3 for proper generation of LOAT and LOAF.

The aim is to generate the LOAT or LOAF or both as coded in the appendices, avoiding blank lists with only headers or no lists altogether. The other items of the TOC, namely, LOT, LOF, LOSH, LOA,

Table 3: Appendix closing for LOAT and LOAF generation—last command before `\end{document}`

Previous appendix(s) has/have	Last appendix has	Command to be used	Comment
No Table & Figure	No Table & Figure	None	LOAT and LOAF not generated
No Table & Figure	Only Table	None	Only LOAT auto generated
No Table & Figure	Only Figure	None	Only LOAF auto generated
No Table & Figure	Table & Figure	None	LOAT and LOAF auto generated
Only Table	Only Table	None	Only LOAT auto generated
Only Table	Only Figure	`\closeappendixtables`	LOAT and LOAF generated
Only Table	No Table or Figure	`\closeappendixtables`	Only LOAT generated
Only Figure	Only Table	`\closeappendixfigures`	LOAT and LOAF generated
Only Figure	Only Figure	None	Only LOAF auto generated
Only Figure	No Table or Figure	`\closeappendixfigures`	Only LOAF generated
Table & Figure	Only Table*	`\closeappendixfigures`	LOAT and LOAF generated
Table & Figure	Only Figure*	`\closeappendixtables`	LOAT and LOAF generated
Table & Figure	No Table or Figure	`\closeappendices`	LOAT and LOAF generated

* The command `\closeappendices` can also be used instead

and LOS, are automatically generated based on the presence or absence of these floats/items coded in the chapters or prefatory material.

13 Additional Information I—Special Commands

13.1 Chapter styles

Two styles, namely, regular- and paper-styled chapters, are generally followed. The regular is a traditional style where the whole thesis/dissertation is considered as a single document where individual chapters exclusively deal with aspects like introduction, literature review, methods, results, discussion or results and discussion, references, and appendices reflecting all studies carried out in the research on these individual chapters. Even though this style produces a consolidated document and is solid in its own merit, which ties all research aspects of the study together in corresponding chapters, a good deal of rewriting will be necessary from the authors if they want to publish the contents as individual peer-reviewed journal articles.

The paper-styled chapters are stand-alone chapters complete with all sections (abstract, introduction, literature review, . . . , references) and are the modern trend. In this style, some amount of repetition among chapters is unavoidable (especially in methods, analysis, and references). However, as the chapters are already in paper-style, it is very easy to format them to suit the requirements of any peer-reviewed journal for submission. It is also possible to have individual chapter references (BibLATEX) or a combined reference chapter (both BibTEX and BibLATEX). With LATEX it is easy to create stand-alone papers with references for submission from a paper-styled disquisition with a combined reference chapter. As outlined earlier, the commands that start these chapters are `\myheading{...}` or `\mypaperheading{2 args}` (section 12.2). It is a good idea to consult the advisor before committing to these styles, for they are different, and substantial rewriting is involved to switch back and forth.

13.2 Advanced options in documentclass

13.2.1 Font size

The general font sizes used with the thesis are 10, 11, and 12 points, and they vary with the selected font. However, for accessibility, only 12 points and above are recommended, and the other sizes can be used for drafts and personal copy. The available options (any of these used) are:

`\documentclass[10pt (or) 11pt (or) 12pt]{ndsu-thesis-2026}`

The default was set as `12pt`.

13.2.2 Auto-numbered, chapter-numbered, and unnumbered styles

The three possible NDSU thesis styles with options included are: (i) Auto-numbered [default option]—where chapters, sections, subsections, and so on will be numbered; (ii) Chapter-numbered [chapternumber]—where only chapters are numbered, while sections, subsections, and so on will not be numbered; and (iii) Unnumbered [nonumber]—where all headings such as chapters, sections, subsections, and so on will not be numbered.

As the default is the numbered style, the chapter-numbered and unnumbered styles were produced by the "chapternumber" and "nonumber" options, respectively as:

`\documentclass[chapternumber (or) nonumber]{ndsu-thesis-2026}`

. The default was the "Auto-numbered" style. These options will have their specific effect on the numbering scheme of the tables and figures.

13.2.3 Paragraph text justification

Based on their preference, students can follow fully-justified (with hyphenated words and word wrapping) or unjustified (no word breaking but right margin ragged, aka left-justified). As the default is justified, the left-justified passages were produced by the `[nojustify]` option. For justified style, nothing needs to be specified. NDSU approves both styles.

13.2.4 Draft and display document frames

You can use the `[draft]` option to place the disquisition into draft mode. In this mode, margin overflows are marked with a heavy black box to draw your attention to them; additionally, images are replaced by a placeholder (fig. 2a). If you import other packages in your disquisition, they may also change their behavior when in draft mode.

The `[showframe]` option (based on `geometry` package) produces a frame around the text area, which can be used to check how the text aligns with the margins (left, right, top, and bottom; see figure above). The illustration alongside displays the result of these options, showing the overflowing text, bottom margin frame, right margin frame, margin notes frames, and the overflow heavy black box. The default behavior is that these options are inactive.

13.2.5 Grids display

While alignment and spacing of elements will be automatically handled by LaTeX, the NDSU's guidelines deviate a bit from the LaTeX engine, especially texts around non-textual elements, viz. equations, tables, and figures. Although the normal behavior of LaTeX introduces a little extra space around non-textual elements, NDSU's guideline dictates a consistent double-spacing that is followed in the regular text, which should be around the non-textual elements. This additional spacing can be corrected by using a −ve `\vspace{...}` command before and after the elements as required (section 13.7 for details).

To visualize better the spacing around textual and non-textual elements, a documentclass `[showgrid]` option was made available (fig. 2b). The grid was displayed underneath the elements covering the body

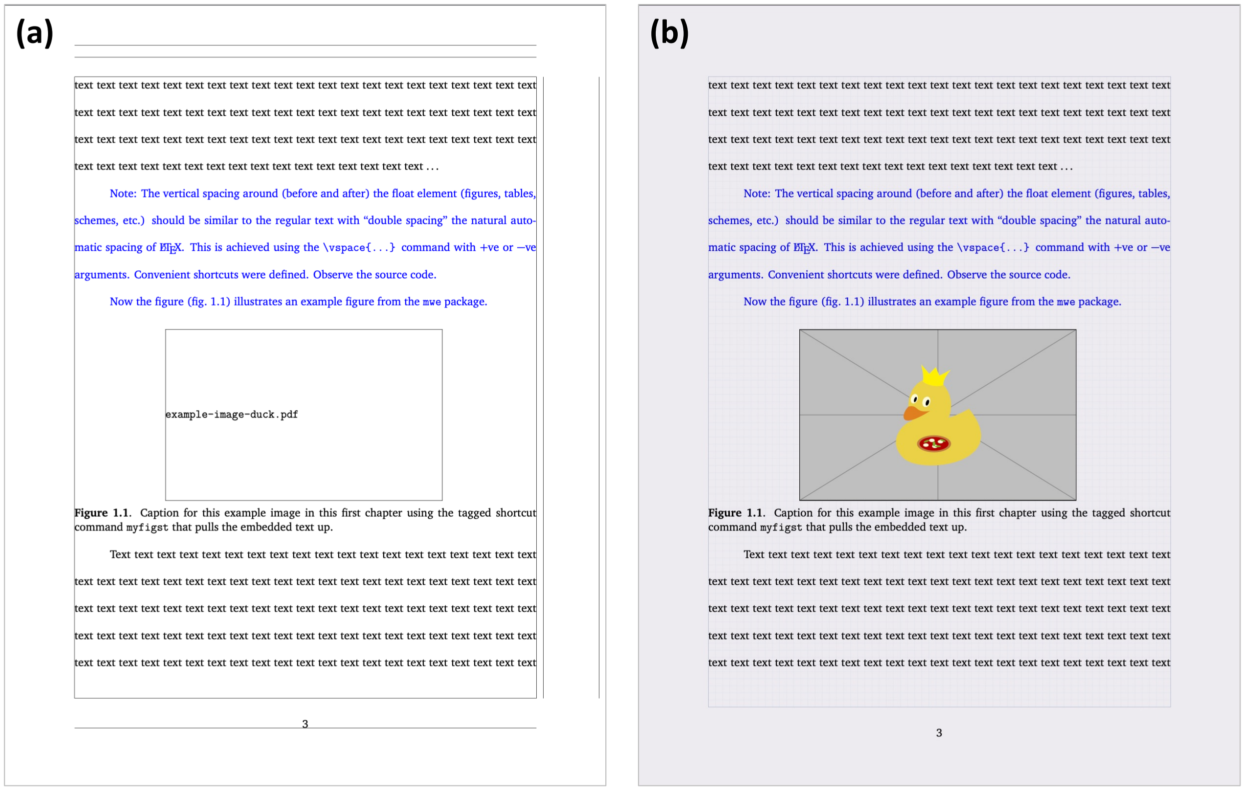


Figure 2: Use of (a) `draft` and `showframe` options in `documentclass` producing image placeholder for quicker processing, document frames, and margin overflows, and (b) use of `showgrid` option displaying grids of 0.1 in squares spacing to help visualize the alignment (vertical and horizontal) concerns of elements.

(`textwidth` $\times$ `text height`), and the lines were spaced at 0.1 in on both vertical and horizontal directions, and the whole page is rendered on a light gray background. It can be seen that the basic text line vertical spacing is about 0.4 in (4 grid lines between the baselines of consecutive lines of text; fig. 2b). This vertical spacing (0.4 in) should be carried throughout the document with some deviation among captions of tables and figures, table data rows, and program listings. After fixing the spacing concerns, using `\vspace{...}` commands, the `showgrid` option should be removed for the final output. By default, this option is inactive.

13.2.6 Fonts

The following font options [`bookman, charis, charter, clearsans, cmbright, firasans, helvet, lxfonts, kurier, libertine, lxfonts, mathdesign, mathptmx, mlmodern, newcent, newcomputermodern, newpx, newtx, opensans, palatino, sansmathfonts, times, tgbonum, tgpagella, tgschola, utopia` are loaded and compatible with the class, and anyone can be used. As the NTTC uses LuaTeX, the fonts supported by `fontspec` package will work. The default was LaTeX computer modern font. Users are urged to check the NDSU-approved fonts and select those that resemble them and use them with appropriate font sizes.

13.3 Line numbers for the whole document

Sometimes using line numbers will be helpful while communicating with the advisor or others, where specific locations of the document can be pointed to. Line numbers are generated using the package `lineno`, which is coded into the class, by the following command:

```
\linenumbers
```

This command can be issued at the beginning or at any point, and numbers will appear in the left margin after the command. Of course, this command should be removed or commented on while finalizing the thesis.

13.4 Whole document text spacing

NDSU mandates double-spacing for the body paragraphs' text. A default double-spacing setting in MS Word produces 23 lines per page while LaTeX `\doublespacing` produces 27 lines. Any of these lines per page defaults in the respective systems is acceptable. To recreate the line spacing of 23 lines per page in LaTeX was reproduced through `\setstretch{1.9}` (applicable to "`mathdesign`" and "`computer modern`"), which was also set as the default in the class, and this spacing is automatically applied to the whole document. For other fonts, the `\setstretch{...}` value should be modified to produce 23 lines/page, and this command should be issued inside the `document` environment.

13.5 Tables: Advanced commands

13.5.1 Table row spacing and fonts

The table contents row spacing can be adjusted, if desired, using `\renewcommand\arraystretch}{decimal}` before `tabular or xltabular` environment inside the table environment. Also, `\singlespacing` may be issued to produce the singlespacing among the rows. For example, a value of 1.75 for the arraystretch will be similar to the double line spacing, and without this command, the row spacing will be single line spacing. Table footnotes can be added through `\multicolumn{...}` command. The footnote font size should be similar to the table contents (see example thesis).

13.5.2 Landscape tables

When a table has more columns of information, the most common solution is the landscape orientation which is achieved through `landscape` environment by enclosing the `table` codes (which may contain other elements) inside `landscape` environment block (between `\begin{landscape}` and `\end{landscape}`). With landscape, usually the placement option will be [p], and the whole width should be set around 1.32 times the `\columnwidth`, or adjusted suitably to leave acceptable margins all around.

13.5.3 Resizing tables for full-width

Sometimes, while fitting more content into a table, the table extends beyond the allowed margins. Therefore, for the best control of tables, especially with more columns, a combination of `\resizebox` (resizing the entire table - mostly for scaling down) and `\tabcolsep` (maintaining the column separation space) works the best. Thus, the command `\resizebox{\columnwidth}{!}{ ...table codes ... }` makes the table span the entire text width of the page. This will expand or shrink the contents of the table to fit the entire width. It should be fine with the fonts shrinking to fit the width, but it will not be when the fonts enlarge (especially when the table is small and has only a few columns). In such situations, the space between the columns can be adjusted using the `\tabcolsep{...}` command, where increased spacing reduces the font size and *vice versa*.

13.5.4 Long tables

There will be a need to create long tables (multiple pages spanning tables) that do not fit into a single page. It should be noted that, unlike regular tables, long table source code involves several components (main caption, running header [abbreviated caption], running footer [usually the word "continued..."], and main footnote). The `xltabular or longtable` packages are capable of producing tagging-compatible long tables. The **Table 3.2** in the example thesis serves as a good example. Also, refer to the "`NDSU-Thesis-Extended.tex`" (link:) for sample code.

13.5.5 Landscape long tables

Sometimes, there will be a need to create long tables (multiple pages spanning tables) in the landscape orientation to accommodate several columns that will not fit in regular paper orientation. This landscape long table can be logically obtained by enclosing the long table codes within `landscape` environment block as described earlier. Combining the `landscape` and `longtable` environments, the landscape long tables can be created logically. Refer to the "`NDSU-Thesis-Extended.tex`" (link:) for an example.

13.6 Figures: Advanced commands

13.6.1 Figures and schemes in separate folder

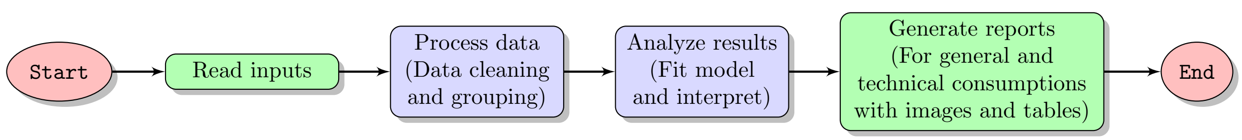
Several images (figures, schemes, graphs, drawings, and pictures) were used while developing a thesis or paper. It will be convenient to store all these images in a subfolder to reduce the clutter. The following command should be issued in the preamble, indicating the name of the subfolder (e.g., `figures`) relative to the main "`tex`" file as:

```
\graphicspath{{./figures/}}
```

The type of image files applicable are: `jpg, pdf, png,` and `eps`. It is also possible to give an absolute path to the images folder in the above command.

13.6.2 Flowchart - tikz package

Flowcharts, schemes, geometrical diagrams, circuit diagrams, and data visualization graphs are common in technical writing. These elements can be created elsewhere and included in the dissertation as an image, or high-quality (vector graphics) can be created using code directly. An example of a schematic flowchart created through Ti*k*Z code is shown below:



Schematic 1: A high-quality schematic flowchart created using the Ti*k*Z package.

The Ti*k*Z package, based on TeX is an excellent and elaborate package (manual having > 1300 pages) that can be used for creating high-quality graphics that serve the needs of any technical documentation (Example Sch. 1). Going through the manual of Ti*k*Z and the gallery will give information on the package capabilities and how they can be used in the dissertation.

13.6.3 Subfigures

As packages dealing with subfigures are not currently tagging-compatible, the subfigures should be generated using other software and utilized as a single figure directly. In the caption and alt-text, the subfigures can be described. In the future, new packages will resolve this issue.

13.7 Spacing adjustment around non-textual elements

Usually, the spacing around the non-textual elements produced by LaTeX will be good and based on typography principles. The environments that create these elements (e.g., tables, figures, schemes, equations) automatically supply an additional space to set the elements apart from the regular text, and this is the expected and correct behavior. However, sometimes additional space will appear above or below these elements, which may be the result of fitting the elements with respect to others in the whole chapter.

The spacing around the non-textual elements can be altered by one or any combination of the following to produce a consistent spacing around the non-textual elements, similar to the double line spacing of the textual—which is an NDSU thesis requirement. To have the "regular" spacing all around various elements, the method is to use the `\vspace{...}` command with −ve or +ve values with appropriate units.

13.7.1 Vertical spacing quick method—shortcuts

It is convenient and quick to follow these steps to ensure uniform spacing throughout the thesis. The ideal target spacing we would like to achieve throughout is 3 grids (0.3 in) gap between the bottom and the top of lines (fig. 3). This spacing is optimized for LaTeX default "`computer modern`" and "`mathdesign`" fonts. Other fonts may have different spacings, but could follow similar strategies (Section 13.4).

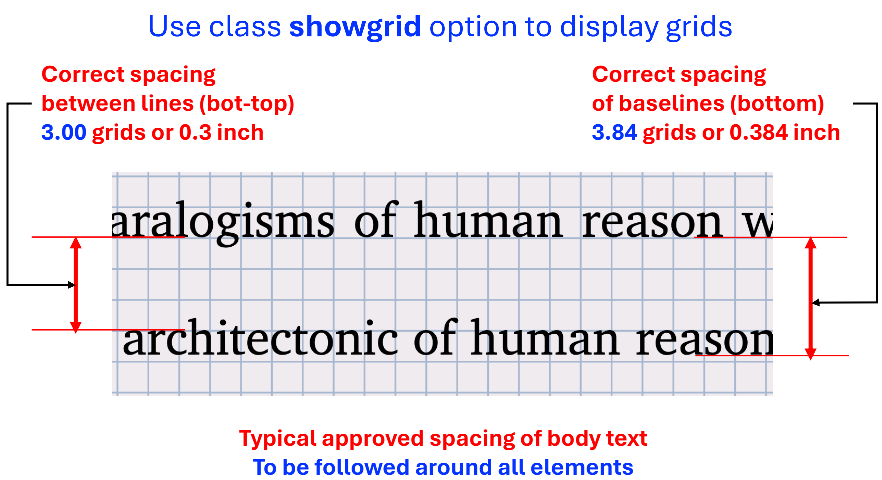


Figure 3: Vertical spacing strategy illustrating the "target/ideal" 3 grids spacing (0.3 in) gap.

- ☞ **Showgrid**: Have a `[showgrid]` option active. Helps with easy spacing visualization. Grid color was made so light that it would not interfere while working, and grids can be removed only for the final output.
- ☞ **Strategy**: As the target gap is 3 grids (fig. 3), it is adequate to work in terms of "grids" (1 grid = 0.1 in). The smallest unit that we need is $\frac{1}{2}$ grid. Most of the time, spacing adjustment means reducing the space by `\vspace{...}` command with −ve values, while increasing the vertical space with +ve values is rare.
- ☞ **Shortcuts**: Therefore, convenient shortcuts with −ve values, such as `\vh`, `\vo`, `\voh`, `\vt`, `\vth`, `\vr`, and `\vrh` that represent "vspace-half-grid" (which is equivalent to `\vspace{-0.05in}`), "vspace-one-grid" [vo], similarly, one-half [voh], two [vt], two-half [vth], three [vr], and three-half [vrh], respectively, were defined in the class.
- ☞ **Around figure**:
  (i) Figure between two paragraphs: This is a common scenario, and the figure is usually placed as requested (not floated to the next page). The additional spacing around the figure can be controlled by issuing vertical spacing shortcuts (above and/or below; outside of figure environment) in the usual manner (e.g., `\vo`, `\voh`, `\vt`, `\vth`, ...).
  (ii) Floated figure embedded in a paragraph: This is rare but occurs. Here, the figure is floated to the next page, usually to the top. The top margin of the figure may be fine, but the bottom will have the additional space. As the logic of the shortcut controls the vertical space around the caption, the third optional argument can be set to adjust the space below the caption.
- ☞ **Around table**:
  (i) Table between two paragraphs: Similar to the figure between two paragraphs, and the shortcuts can be used above and/or below on the outside of the table environment.
  (ii) Floated table embedded in a paragraph: In principle, similar to the floated figure embedded in a paragraph. The table got floated to the next page, and the adjustment `\vspace` command should be

issued inside the table environment, usually near the end after the footnote code, if present. Shortcuts (e.g., \vo, \voh, \vt, \vth, ...) can be used as tables are directly coded.

☞ **Headings on top of page**: Headings, such as `section, subsection` and so on, when they happen to be on the top of the page, will have more than a grid space above them from the top 1 in margin, and need to be addressed. This is again the situation of the element "getting floated" to the next page for want of necessary space to accommodate them in the previous page. To remove the gap, the shortcut `\mysectop` can be used before the heading code. It should be noted that any simple vspace commands will not work as the method requires clearing the page.

☞ **Other shortcuts**: These include (i) `\vs{}` (defined as `\vspace*{#1in}`; "vspace starred") for the non-discarding starred version of vspace for guaranteed moving of items vertically, and (ii) `\vc{}` (defined as `\clearpage\vspace*{#1in}`; "vspace clearpage" with starred vspace) for moving items to align with the top margin. These shortcuts take a numerical value, hence versatile, and the default unit is inch (works well with grid spacing).

☞ **Regular way**: The shortcuts discussed so far are made for convenience. They use just the three regular commands i.e. `\vspace{}`, `\vspace*{}`, and `\clearpage`, along with the inch unit. Therefore, in the regular way, these commands can be used instead to achieve the same result.

13.7.2 Vertical spacing—other information

- An additional document named "ndsu-thesis-vertical-spacing" that addresses this is available in the package (link:) .
- The issue becomes apparent when the full page float is taken to the subsequent page and "vertically" centered. This breaks the rule of a "1-inch" top margin. The solution in simpler terms for this is: (i) Give a `\newpage` command before the float environment, (ii) use placement specifier `[h!]` or `[H]` not `[p]`, and (iii) the blank space before the float (in the previous page) is okay (eventhough akward as this is unavoidable) as this larger float could not have been accommodated there in the first place.
- The blank line coded, usually left between paragraphs, might create additional space before the element (e.g.,`equation`, `align`), and that can be removed to reduce the space above the element. For equations, the defined shortcuts can be used to produce the correct vertical spacing (Section 12.8.1).
- Proper use of the vertical spacing `\vspace{...}` command with negative spacing (e.g., `\vspace{-3ex}`) can able to correct the blank space above the element. This can also be used when a blank line was issued to separate the regular text from the element. Positive vertical space can also be issued as needed.
- In figures and schemes, the space above the float as well as the space above and below the caption can be controlled by using the optional arguments of the `mytfig, mytfigls, mytsch, mytschls, mytfigap, mytfiglsap, mytschap` and `mytschlsap` commands (Section 12.5.1). This may be required only for necessary adjustments, as the default (without option) will work well in most cases.

13.8 Reference handling using BibLaTeX

The BibLaTeX package provides advanced bibliographic facilities for use with LaTeX. A good working knowledge of LaTeX should be sufficient to design new bibliography and citation styles using this system. The BibLaTeX works with the backend (program) "biber", which is used to process the bibliography data files and then performs all sorting, label generation, and many more operations. This package also supports subdivided bibliographies, multiple bibliographies within one document, customizable sorting, multiple bibliographies with different sorting, customizable labels, and bibliographies may be subdivided into parts and/or segmented by topics. Users are urged to refer to the package documentation for various features (`https://ctan.org/pkg/biblatex?lang=en`).

13.8.1 Commands, cite, bibliography generation, and file handling

The basic commands that invoke the BibLaTeX system, which have to be issued in the preamble, are:

```
\usepackage[style=apa,natbib=true,backend=biber]{biblatex}
\addbibresource{name-of-bib-file.bib}
```

In the above BibLaTeX command's option, the bibliography information processing program "biber" was used as a backend program. The options also load "apa" style and "natbib" handling (allowing the `\citep{...}` and `\citet{...}` commands in BibLaTeX) as an example. The documentation and other resources (`http://tug.ctan.org/info/biblatex-cheatsheet/biblatex-cheatsheet.pdf`) may be referred to for common options and details of the package. The compatible styles used with BibLaTeX are: `numeric, numeric-comp, alphabetic, authoryear, authoryear-icomp, authortitle, verbose, reading, draft, apa, chem-acs, chem-angew, chem-biochem, chem-rsc, ieee, mla, musuos, nature, nejm, phys, science,, oscola`, and so on (new styles are made available with every update). Users can use an appropriate style to match their specialization style guide.

Note: Only with numerical style bibliography, such as `ieee`, `nature`, `numeric`, `numeric-comp`, `...`, to have the numbers in the reference listing left-aligned (especially > 10 items), it is best to copy this code and paste it after the `\addbibresource{...}` command in the thesis source (*.tex) file (around line 15).

```
\makeatletter % Make @ available for internal macros
    \defbibenvironment{bibliography}
    {\list
        {\printtext[labelnumberwidth]{%
        \printfield{labelprefix}%
        \printfield{labelnumber}}
        }%
        {\setlength{\labelwidth}{\labelnumberwidth}%
        \setlength{\leftmargin}{\labelwidth}%
        \setlength{\labelsep}{\biblabelsep}%
        \addtolength{\leftmargin}{\labelsep}%
        \setlength{\itemsep}{\bibitemsep}%
        \setlength{\parsep}{\bibparsep}}%
        \renewcommand*{\makelabel}[1]{##1}
        }
    {\endlist}
    {\item}
\makeatother
```

With the package and bib file(s) loaded and processed, the reference listing can be generated anywhere in the document by issuing:

```
\printbibliography[heading=bibintoc,title=REFERENCES]
```

The options "`heading=bibintoc`" make an unnumbered chapter and include the heading in the TOC, and "`title=REFERENCES`" changes the default title from BIBLIOGRAPHY to REFERENCES.
The `\printbibliography` command, when issued at the end of the chapters, will create a "combined" REFERENCES chapter.

13.8.2 Automatic individual (multiple) and whole document reference listing using BibLaTeX

Sometimes it is desired to have a bibliography listing in every chapter, as the last unnumbered section, especially with the paper-style chapters. These individual chapter reference listings can be invoked by the option `[chapterrefs]` in the documentclass. Individual chapters' bibliography can be easily processed using BibLaTeX rather than BibTeX through `refsection` environment.

New commands such as `\checkBeginRefsection` and `\checkEndRefsection` can be considered as an environment that immediately follows each chapter title and encloses the chapter contents (similar to `refsection` environment)—as shown subsequently. At the end, usually before appendices, the `\checkMakeCombinedReferences` command was once issued.

```
\mypaperheading{...}{...} % chapter n
\checkBeginRefsection
```

...Chapter's text starting with abstract, sections/subsections, and so on,
with citations using \citep{...} and \citet{...} or other citation commands ...

```
\checkEndRefsection
\mypaperheading{...}{...} %chapter n+1
\checkBeginRefsection
```

...Chapter's text starting with abstract, sections/subsections, and so on,
with citations using \citep{...} and \citet{...} or other citation commands ...

```
\checkEndRefsection
```

...Other chapters' codes follow a similar arrangement

```
\checkMakeCombinedReferences % Given at the end of chapters
```

...Appendices codes ...

A combination of these commands—automatically checks the desired type of reference listing—and produces the individual chapter reference listings (with `[chapterrefs]` documentclass option) or the whole document unnumbered reference chapter (without the aforementioned option - default). Following these automatic commands of BibLATEX will be the most efficient way of handling the bibliography. Examples of the usage of these commands will be seen in the included "`Sample-thesis-IncludeOnly`" folder files. Needless to mention, these are convenient commands developed for the class, and the desired behavior can also be achieved by usual direct commands.

13.9 Reference handling using BibTeX

13.9.1 BibTeX

The compatible styles (`*.bst`) with `natbib` and NDSU class that work with standard LATEX installation are: `abbrvnat, agsm, agu, apalike, apalike2, authordate1, authordate3, cell, chicago, chicagoa, dcu, dinat, IEEEtran (family; numerical styles), kluwer, plainnat, rusnat, unsrtnat`, and so on (new styles are made available with every update). For other styles, users can able to download the specific style (`*.bst`) files and have them in the local folder. As `natbib` is an optional bibliography system, it was not coded in the class, and to use `natbib`, the following code should be in the preamble:

```
\usepackage[sort&compress]{natbib}
\citestyle{arms} % agms, agu, arms, egu, cospar, dcu, kluwer, plain, nature
```

It is convenient to load the `natbib` package with minimal options, as shown above, and choose predefined `\citestyle{option}` options producing several styles defined in the package.

13.9.2 Bibliography generation and files handling

The two basic commands that are required to implement BibTeX system are:

```
\bibliographystyle{style} % See list of styles (Sec.11.1.1)
\bibliography{name-of-bib-file}
```

However, a single new command "`\makebib`" (direct shortcut with no arguments) was coded, replacing the above commands, setting up the bibliography style and *.bib file arguments as:

```
\newcommand\makebib{\biblio{style}{name-of-bib-file}}
```

The above command is conveniently placed at the beginning of the preamble, where the users replace the style and bibliography arguments as inputs. Then, simply issuing the `\makebib` will generate the references listing based on the style and bib file input in the above command. The "style" of bibliography (`*.bst`) entries (typically `plainnat` or `apalike`), is controlled by the first argument; the user is referred to the BibTeX manual for formatting details and other available styles, such as those provided by the peer-reviewed journals related to the specialization. The "name-of-bib-file" used in the second argument must be the same as the name of the bibliography (`*.bib`) file, but with the extension removed. Once correct citation commands (`\citep{...}` and/or `\citet{...}`) are issued following CWYW, the citation with proper reference number or entry will appear in the text and listings in the proper style (based on *.bst) will be generated. The above shortcut generates an unnumbered chapter with the title REFERENCES (accepted by NDSU) and also a corresponding TOC entry.

These commands (or equivalent commands if the user uses a different bibliography management system) are optional but are required if the disquisition includes references. The basic bibliography citation command is `\cite{...}`, and that works as well.

13.10 Specialization specific bibliography—Examples of ASABE and IEEE

The bibliography styles to be followed in the thesis/dissertation will be based on the parent department and the major technical society the department subscribes to, in general. For example, the NDUS's "Agricultural and Biosystems Engineering" department's major technical society is the American Society of Agricultural and Biological Engineers (ASABE), and the NDSU's "Electrical and Computer Engineering" department's technical society is "The Institute of Electrical and Electronics Engineers (IEEE)." The two ways of making the bibliography are to use BibLaTeX or BibTeX. The ASABE style journal references can be reproduced using `\usepackage[style=apa, natbib=true] {biblatex}`.

For `IEEEtran` styles, with BibLaTeX any of the following can be used (with increasing functionality). With BibTeX `*.bst` files and `\citestyle{...}` commands are available to create the style of reference followed in various departments (Section 13.8 for details).

```
\usepackage[style=ieee,backend=biber]{biblatex}
\usepackage[style=ieee,natbib=true,backend=biber]{biblatex}
\usepackage[style=ieee,natbib=true,citestyle=numeric-comp,backend=biber]{biblatex}
```

13.11 Bibliography compilation issues

The common issues [and proposed solution] while working with bibliographies, especially in combination with `natbib` include:

- The basic principle usually gets violated is that the LaTeX inputs (`tex, bib, or others`) require only the ASCII characters. Anything that disregards this principle will have issues during compilation. Presence of extraneous characters (unprintable and formatting) in the `*.bib` file, which comes while copying a formatted non-ASCII text, stalls the compilation. Command line or terminal commands or online tools can be used to identify the non-ASCII characters (e.g., `https://pages.cs.wisc.edu/~markm/ascii.html`, `https://onlineasciitools.com/validate-ascii`) [remove non-ASCII text from `*.bib` file]
- BibTeX contains several types of entries (e.g., article, book, misc, etc) and each has its own "Required" and "Optional" fields. Not filling the required field will throw an error [familiarize with these fields (`https://en.wikipedia.org/wiki/BibTeX`) and ensure supplying all the required fields or change to a suitable different entry that is less restrictive (e.g., misc)]
- Another most common issue with bibliography entries is "duplicate entries." These are entries with repeated citation "keys" (non-unique) and stop the compilation with a clear error message [check and remove the duplicate/multiple entries]
- The bib files should not contain regular text, such as notes. All items should belong to allowed database entries and fields [check and remove the text entries or comment them out using %]

- Proper title case will not appear in the output (e.g., proper nouns like country names) even though things may seem right in `*.bib` file [use additional curly braces {...} to enclose the desired title case items or the letters]. This happens especially with the title entries of the `*.bib`, but the journal entry will be fine.
- Accented characters, usually found in authors' names, might cause issues [correct use of codes to produce accented characters should be used and it is a good idea to enclose them with additional braces (e.g., é, ò, ê, ö, ç, ñ, ł, Ł, æ, å, through `{\'e}`, `{\`o}`, `{\^e}`, `{\"o}`, `{\c c}`, `{\~n}`, `{\l}`, `{\L}`, `{\ae}`, and `{\aa}`)]
- Proper italics will not show in the output (*Scientific names* of animals and plants; e.g., *Homo sapiens, Zea mays*) [use `\emph{...}` around the entries in the `*.bib` file]
- Proper formatting of URLs with wrapping content and clickable links does not appear [use `\url{...}` command in the `*.bib` files entries that have the URLs]
- Outputs not formatted according to the thesis/journal requirement [select the correct BibLaTeX style option or `bst` file that matches most of the requirements and modify the code to suit our requirement might sometimes require—most often only a few alterations are needed and have the file in the same folder as `*.bib`]
- New bibliography styles that are not listed (Section 13.8.1) might not run properly, mostly because of not having the corresponding BibLaTeX style or `*.bst` in the system [download style files and have it in the system/local folder]
- Some numerical styles (e.g., IEEEtran family) will not work when the `natbib` was set to author-year format [optional arguments of `natbib` package or arguments of `\citestyle{}` command should be made compatible to the numerical styles (e.g., "numbers" or "plain" in their respective commands)—so a combination of chosen `*.bst` and `citestyle option` is the key to create the references listing in the correct format]
- Numerical bibliography styles reference listings with left-aligned numbers. [See the note on the Section 13.8.1]
- While trying different styles, compilation stops even for compatible styles [this is a common phenomenon and could be solved with an understanding of how BibTeX and LaTeX resolves references by creating several auxiliary files (e.g., *.aux, *.toc, *.log, *.out) including *.bbl—as contents of these files were referred from unsuccessful compilation, removing all these files leaving the source (`*.tex`) and recompiling with the compatible style and options will restore and regenerate the output. Thus, starting with only a clean source will work and regenerate the necessary files, including the output; this strategy will work in other situations of compiling issues as well—however, it is not always necessary].

13.12 Proper development of bibliography bib file

- Bib files should not be considered a dumping ground for bibliography codes that were usually not looked at for correctness, unlike the regular document source codes. This sometimes proves to be a serious mistake and stops compilation or produces strange outputs or major errors.
- The bib file, therefore, should be developed step-by-step carefully, and each entry should be checked for correctness (only with ASCII characters and allowed equivalent LaTeX commands for non-ASCII characters) and appropriateness (required and optional fields in bibliography entry).
- Bib files are automatically generated by other reference management software (e.g., Zotero) for a given set of references, while they are helpful and provide a good starting point, they are sometimes sources of compilation errors. This software may include "abstract" fields as well in the bib file, which are long and become sources of several unintended non-ASCII characters. These superfluous fields may be deleted so that the bib file is compact. Some of these software may include modern entries and fields that may be recognized by BibLaTeX but not included in BibTeX. Though it may be possible to create simpler and cleaner bib files, it is a good idea to check the generated bib file "line-by-line" for correctness.
- It was found that "Google Scholar" generated BibTeX entries worked well in developing the bib file. These entries usually generate logical naming of the keys and valid LaTeX commands for non-ASCII characters

(especially international accented characters). Even in these entries, checking the bib entries is always a good idea (e.g., use of capital letters and emphasis).

- Utilize the BibTeX-Tidy web tool (`https://flamingtempura.github.io/bibtex-tidy/index.html`) to clean, sort, remove duplicates, remove fields, remove comments, etc., and make a tidy bib file.
- Common errors include:
  - ➔ Extraneous non-ASCII characters (Section 13.11 first bullet);
  - ➔ % sign - which is an innocent symbol, but with LaTeX anything after will be considered as a comment, so to produce % we need to use \%;
  - ➔ On a similar note, we have a set of reserved symbols (&, %, $, {, }, , _, #, and so on) that carry a special meaning and should not be used (without "\" preceding; e.g., \&, \%, \$, and so on) loosely and expect a logical outcome;
  - ➔ Every field (author, year, journal, etc.) should end with a comma (but the last one) - in a good bib file, just remove one comma and see the effect, which will teach us how a small thing will wreck the whole compilation;
  - ➔ Unconventional fields (though created by other software) should be replaced by the correct ones (Wikipedia BibTeX), for example, date instead of the year will not work;
  - ➔ Special symbols like degrees, copyright, and others commonly used in other systems can be coded in LaTeX as regular commands - and that command version of these symbols should be used in a bib or tex file;
  - ➔ Duplicate entries;
  - ➔ Unmatched braces;
  - ➔ Several types of BibTeX entries (`https://en.wikipedia.org/wiki/BibTeX#Basic_structure`; e.g., `article, book, inproceedings, masterthesis, phdthesis, misc`, and so on) with their specific *Required fields* and *Optional fields*. Missing a required field (though generated by other resources) will be a mistake and will suspend compilation. The entry *misc* does not contain any required field and can be used to fit odd entries;
  - ➔ The foreign accented characters (check the LaTeX cheat sheet and learn how to code them - all start with \ followed by some ASCII symbols or characters and the input character that receives the accent (Section 13.11). As a practice, try to have these accented characters in the tex file. If it runs there, then that can be inserted into the bib file.
- While building your bib file, add a couple of entries and check how they compile. Don't go for the whole 120 entries (just to throw a good number).
- To visualize the reference listing of all the entries of the `*.bib`file issue the `\nocite{*}` command. After the correctness of the entries is checked and fixed, the command may be removed, and the listing of only cited references will appear—nothing more, nothing less.
- Further, remember the use of enclosing capital letters, especially in titles, for example, NDSU, Argentina to get the capitals right from bib files. Similarly, the use of `\emph{...}` for technical names in bib files for italicized output (e.g., *Panthera leo* for lion) - otherwise they come out in normal font.
- It is a good idea, while working with bib entries or in the main text, the liberal use of the `comment` environment, from the `comment` package, in the main text helps to check problematic bib entries and also produces efficient compilation.

13.13 Chemical symbols

Chemical symbols and chemical equations can be coded easily in a natural manner using the `\ce{...}` command using the `mhchem` package—rather than using the math mode. The following chemicals: H_2O, H_2SO_4, $CrO_4{}^{2-}$, $[AgCl_2]^-$, $(NH_4)_2S$, ${}^{227}_{90}Th^+$, and $KCr(SO_4)_2 \cdot 12\,H_2O$ were coded through: `\ce{H2O}`, `\ce{H2SO4}`, `\ce{CrO4^2-}`, `\ce{[AgCl2]-}`, `\ce{(NH4)2S}`, `\ce{^{227}_{90}Th+}`, and `\ce{KCr(SO4)2 * 12 H2O}`, respectively.

$$\verb|\ce{A + B ->[a] C}| \quad \text{gives} \quad \mathrm{A} + \mathrm{B} \xrightarrow{\mathrm{a}} \mathrm{C}$$
$$\verb|\ce{N2 + 3 H2 -> 2 NH3}| \quad \text{gives} \quad \mathrm{N_2} + 3\,\mathrm{H_2} \longrightarrow 2\,\mathrm{NH_3}$$

Refer to `mhchem` documentation for more options and details.

13.14 Individual or combine reference listing

The two types of reference listings are available: (i) in individual chapters as a section, and (ii) a combined reference listing for all chapters as a combined standalone unnumbered chapter. Having the option `[chapterrefs]` produces individual chapter references, and not having the option (default) produces the combined reference listing.

13.15 Annotation commands

While developing the dissertation, the text undergoes several revisions, and suggestions will be provided by the advisor and colleagues. To make suggestions as well as to present the carried out revisions, colored annotations will be helpful to draw users' attention quickly. Therefore, special annotation commands for highlighting, new text, deleted text, replaced text, and notes were defined in the class. These annotation features can be used by the student and the advisor while reviewing the dissertation draft. The `ulem` and `todonotes` packages were used to develop these commands, and their documentation may be referred to for customization. All the annotations can be searched and deleted before submission, and these processes can be even automated by search expressions (e.g., regular expressions). The annotation commands with usage are shown subsequently:

`\hl{Highlight}` gives: Highlight. This will be regular text.
`\nt{Test new text.}` gives: Test new text. This will be regular text.
`\dt{Deleted text.}` gives: ~~Deleted text.~~ This will be regular text.
`\rt{The text to be deleted}{Which will be replaced by this!}` gives:
~~The text to be deleted~~Which will be replaced by this! This will be a regular text again.

While using the above annotation commands, except for `\nt{...}`, enclosing a cited reference commands (`\citep{...}` or `\citet{...}`) use `\mbox {...}` around the cited references. For example, `\dt{...text...\mbox{\citep{daly2010natural}} ...text...}` gives: ~~...text... (Daly, 2010) ...text...~~

`\notes{To Do notes - for interactive communication!}` (also the shortcut `\td{...}`) gives:

To Do notes - for interactive communication!

13.16 Clever reference—cross-referencing items and labels

Referring to items automatically using the defined labels is a common activity in LaTeX and is called cross-referencing. Although there are basic commands available to refer (e.g., fig. `\ref{label}`), the use of `cleveref` package is an efficient way to achieve this task. This package enhances LaTeX's cross-referencing to automatically detect the "type" of the item cross-referenced (e.g., equation, section, tables, figures, schemes, etc.) based on the context of the cross-reference. This means a single command of `\cref{label}` or `\Cref{label}` with the label will produce the correct output (e.g., fig. 1.1, eq. 3, Figure 1.1, Equation 3, etc.). Refer to this package for more details and customization. However, `cleveref` commands will not work with the appendix tables and appendix figures only, where the basic commands (`\ref{...}`) should be used.

13.17 Chapters as individual files

When the length of chapters gets long, it will be better managed into individual `*.tex` files. Then the thesis file will become a collection of such individual files and will be highly compact. The individual chapters are coded using either of these commands:

```
\input{filename.tex}
\include{filename}
```

Nevertheless, it would be more efficient to utilize a single file, as the management and compilation processes would be streamlined. It is important to note that individual component files cannot be compiled because they lack the necessary preamble. The main file that invokes all the component TeX files and includes the preamble will only compile in the conventional manner. Therefore, the recommendation to use a single source code that can be compiled directly is justified.

The use of "block comment environment"—keeps only the current item (chapter or paragraph) in focus, and "temporary ending commands"—ending the document at the point of the command. An example of a block comment covering Chapters 1 and 2 is:

```
\begin{comment}
        ...chapter 1 code ...not compiled
        ...chapter 2 code ...not compiled
\end{comment}
```

An example of a temporary ending with reference listing produced before Chapter 4 is:

```
      ...chapter 3 code ...compiled
\tendblt % issued before the following command
\checkEndRefsection % Don't delete this!
```

13.18 Temporary ending

While reviewing/revising a large document, it will be efficient to compile only the chapter/section that is currently being worked on. The temporary ending command coded in the class, when issued, will generate a shorter document while the rest of the document will be ignored (table 4). As this command only makes a temporary ending, all the source codes from the beginning until the command will appear in the output. To focus and work on a specific chapter/section/paragraph and so on, a combination of the "block comments" environment and "temporary ending" should be used. Thus, using this method, it is possible to work on a single paragraph of source text.

Table 4: Temporary ending commands for different document scenarios and their usage

Document scenario	Command	Outcomes
Single/multiple files	`\tend`	Ends document - should not have `[chapterrefs]` option in documentclass and no reference listing produced
Single/multiple files, BibTeX & who. refs.*	`\tendbt`	Ends document - makes reference listing using BibTeX style
Single/multiple files, BibLATEX & who. refs.*	`\tendblt`	Ends document - makes reference listing using BibLATEX resource, produce chapter individual reference listing with `[chapterrefs]` option otherwise a single whole reference so far

* who. refs. - whole document single reference chapter.

Since we deal with source codes having different features (BibTeX *vs* BibLATEX, individual chapter *vs* whole document reference, single *vs* multiple source files), based on the users' choice, several temporary ending commands were developed (table 4) with usage. Based on the type, these commands will end the document with or without a reference listing. These commands have instructions to appropriately close the respective environments (e.g., `refsection, document`), and users should ensure that the respective environment is active before these commands. It should be noted that the temporary ending commands for BibTeX and BibLATEX should not be applied interchangeably.

13.19 Defining and using specific commands, environments, and packages

As it is not possible to write a class to satisfy the specific requirements of several departments of NDSU, most of the major features as outlined in this document were coded into the NDSU class, and the users can add necessary features in their source code specific to their requirements. This approach gives flexibility, making the class compact and useful. Mathematics, physics, chemistry, literature, etc., departments may use specific symbols and environments that other departments (e.g., agriculture) do not use for their thesis. This means that based on the requirement, if the packages are not already loaded in the class, the necessary packages can be loaded in the source files, and they work with the class. Another aspect is the bibliography style and management, which varies with different departments. To deal with this, the necessary style files (`*.bst`) should be kept in the same folder as `*.tex` or the appropriate path specified in the source code.

Specific packages (e.g., bibliography management), new commands (shortcuts), and new environments (e.g., theorem, proof, etc.) can be included in the preamble. These specific items are best left to individual users, as others may not need this—hence they are not included in the class deliberately.

13.20 Thesis to journal article conversion for submission

A few adjustments are necessary to convert the thesis chapters into journal articles. The commands that are specific to the class (e.g., `\mytfig{}`, `\mytsch{}`, etc.) will not be known to the journal LaTeX template. When the corresponding `newcommand` codes (made of basic commands), as shown in Table 5, against the class shortcut below were inserted in the template preamble for the source to compile. As the shortcuts are convenient and reduce the amount of code typed (Don't repeat yourself—DRY method), it will be better to insert the preamble and retain the thesis code than to replace it with basic commands at all occurrences.

Table 5: Converting thesis chapters to journal articles—preamble information to be added to journal template source code.

Shortcut example commands in thesis (Section 12.5.1)	**Command to be inserted in the preamble for journal articles** (Alternte text not shown)

`\mytfig[-0.1in]`
`{h!}{1.0}{fig1.png}`
`[-0.15in]`
`{Regular figure caption}`
`[-0.2in]`
`{Alternate text for figure}`
`{fig:1}`
	`\NewDocumentCommand{\mytfig}{O{0.1in} m m m O{0.1in} m O{-0.1in} m m}{`
	`\vspace{#1}`
	`\begin{figure}[#2]`
	`    \centering`
	`    \captionsetup{aboveskip = #5 plus 2pt minus 2pt, belowskip = 0pt}`
	`    \includegraphics[width=#3\textwidth,alt={#8}]{#4}`
	`    \caption{#6}`
	`    \label{#9}`
	`    \vspace{#7}`
	`\end{figure}`
	`}`

`\mytfigls[-6pt]`
`{p}{1.32}{fig2.pdf}`
`[-0.15in]`
`{Landscape figure caption}`
`[-0.2in]`
`{Alternate text for figure}`
`{figls:2}`
	`\NewDocumentCommand{\mytfig}{O{0.1in} m m m O{0.1in} m O{-0.1in} m m}{`
	`\vspace{#1}`
	`\begin{landscape}`
	`\begin{figure}[#2]`
	`    \centering`
	`    \captionsetup{aboveskip = #5 plus 2pt minus 2pt, belowskip = 0pt}`
	`    \includegraphics[width=#3\textwidth,alt={#8}]{#4}`
	`    \caption{#6}`
	`    \label{#9}`
	`    \vspace{#7}`
	`\end{figure}`
	`\end{landscape}`
	`}`

Also to be followed with scheme shortcuts when employed.

The compatibility issue will not be there with the tables, for no shortcuts were defined in the class, and all table codes used the basic commands from the respective packages. Also, the heading commands (`\myheading{}` and `\mypaperheading{}`) are specific to the thesis class (Section 12.2) and will not generally work with the journal template. However, their different arguments can be extracted and used in the commands specific to the journal template. Similarly, the code from the appendices, such as `appendixtable` and `appendixfigure` environments, should be replaced by the basic `table` and `figure` environments, respectively, in the article template. In general, a paper will not have schematics, and those are usually coded as figures. Needless to mention, the NDSU class-specific front matter commands that may not appear in individual chapters, such as **`author, date, progdeptchoice, department, cchair, cmember(a-d), approvaldate, approver, listofabbreviations, listofsymbols`**, and so on, will not work in the journal template; but the content of the arguments can be copied and utilized in the appropriate journal template commands.

It may also be necessary to include the missing packages in the template, which were used in the thesis class. The missing packages can be identified from the "`Undefined control sequence`" error messages that stop the compilation (e.g., using `\cref{}` without loading the `cleveref` package). Of course, the template-specific commands should be used while developing journal articles. With these adjustments, the thesis code easily transitions to the journal articles. It is hoped that with the working knowledge of LaTeX previously obtained and/or gathered thus far, the students will be able to handle what is required to convert the thesis chapters into journal articles' source code utilizing the specific journal templates.

13.21 Some shortcuts

Some of the shortcuts and the outputs defined and made available through the NTTC are shown in Table 6.

Table 6: Some of the shortcults made available through NTTC

Command, e.g.	Output	Command, e.g.	Output	Command, e.g.	Output
`\pt{...}`, `\pt{92}`	92 %	`\dct{...}`, `\dct{100}`	100 °C	`\mlt{...}`, `\mlt{20}`	20 µL
`\pr{...}`, `\pr{62.8}`	62.8 %	`\dgt{...}`, `\dgt{114}`	114 °C	`\LX`	LaTeX
`\unit{\USD}`	$	`\unit{\USD}{78}`	$78	`\SU{24}{AnyUnit}`	24 AnyUnit
`\tbp`	•	`\tb`	\	`\tcb{text}`	text
`\tcm{text}`	text	`\cmd{myspace}`	`\myspace`	`\cmdd{myspace}`	`\myspace{...}`
$math `\te{text}`$	*math*text	$D_`\mt{radius}`$	D_{radius}	`\td{...}`	Todo command
`\dsf`	See below	`\repText[20]`	Repeat "text" 20X	`\repInText{my text}{32}`	Repeat "my text" 32X
`\mysectop`	See below	`\mytop`	See below	`\vc{...}`	See below
`\vs{...}`	See below	`ndsugreen`	Color defined ■	`ndsuyellow`	Color defined ■

`\vh`, `\vo`, `\voh`, `\vt`, `\vth`, `\vr`, and `\vrh` are negative vspace commands for lifting items, and the values are -0.05in to -0.35in in steps of -0.05in. `\pvh`, `\pvo`, `\pvoh`, `\pvt`, `\pvth`, `\pvr`, and `\pvrh` similar values and units but positive vspace commands for lowering items. `\mysectop` and `\mytop` are for aligning headings (secs, subsecs, etc.) or other elements (tables, figures, etc.) to the top margin, if they are the first element of the page.

This was completed. Definitions need to be done!

Done so far!

13.22 Accessibility check using Ally checker tool

The template (`ndsu-example-tagged.tex`), derived from NTTC, generates a tagged PDF that adheres to standards and encompasses all necessary properties, including title, author, language, keywords, and bookmarks. This can be verified by examining the "Document Properties" section of the free "Adobe Reader" software (fig. 4).

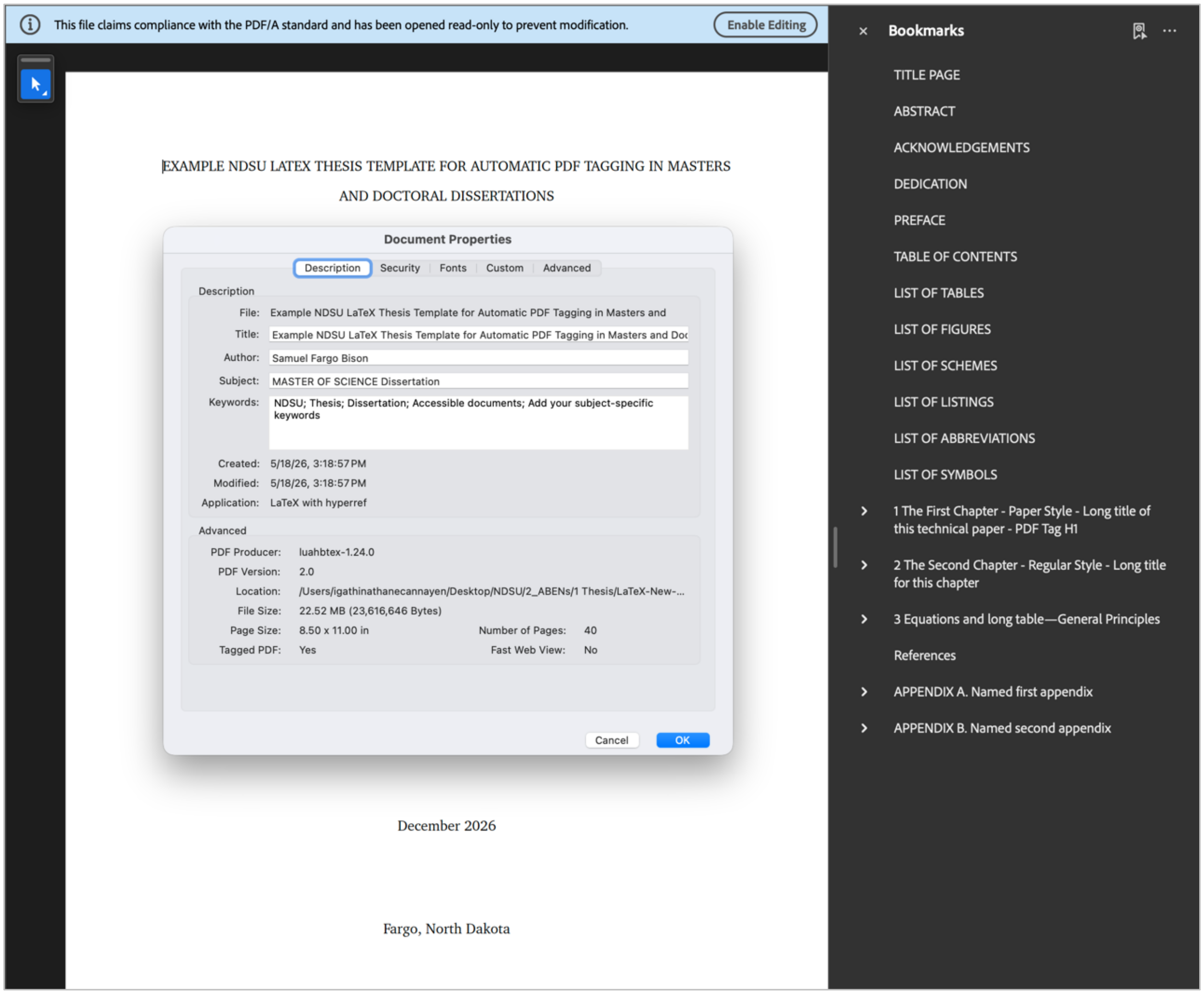


Figure 4: Ally accessibility score for the tagged thesis template example.

The Ally accessibility checker score of 98% is achieved for the example thesis created using the NTTC template, as illustrated in Figure 5. It is noteworthy that there exists a compatibility issue between the PDF 2.0 and the PDF 1.7 versions of the Ally accessibility checker. However, Ally provides an effortless solution through the "Set PDF Title" text box (blue box in fig. 5). Similarly, other tools, such as AAP, have methods to fix this and more. Upon re-entering the title, the recalculated score is found to reach the perfect 100% (fig. 5 bottom-left inset).

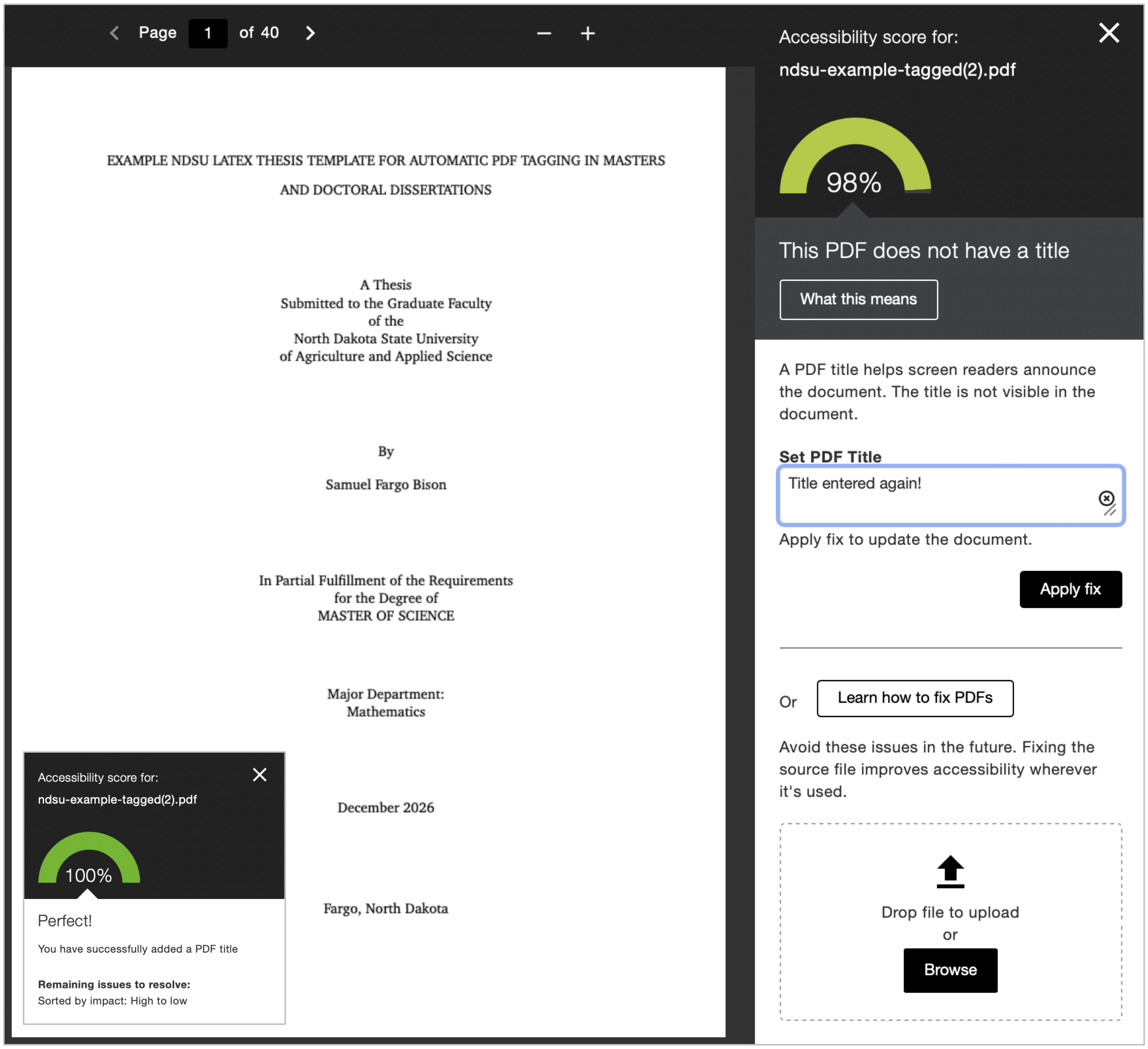


Figure 5: Ally accessibility score for the tagged thesis template example and the inset showing the score after fixing the missing title.

Other third-party accessibility checker tools are anticipated to generate a distinct set of warnings and messages. Novel methods are to be developed, irrespective of the system that generates the thesis, to address those messages. We are collectively learning as we go during the initial phase of "accessible documentation."

14 Additional Information II—Some Tips For Customization

14.1 General suggestions

NDSU Graduate School Formatting Guidelines (link: Graduate School Formatting Guidelines) should be adhered to, and the guidelines to be referred to for various aspects of developing the work. Following are some of the general suggestions while developing the thesis/dissertation:

- All content-related decisions should be made by the student, advisor, and committee, and should follow any rules or conventions established within your program, department, or field.
- Students are highly encouraged to follow the prevalent style manual of the discipline (e.g., MLA, APA, Chicago, IEEE, etc.) while formatting the thesis (especially formatting of tables, figures, and other non-textual elements). In the instances where the Graduate School guidelines contradict the style manual for your discipline, the former takes precedence. However, if a generic style is applied consistently throughout all items, it will also be approved. During format review, a consistent application of one style is accepted. In short, consistency is KEY.
- Paper or regular thesis/dissertation chapters (two possible styles) should follow NDSU format guidelines consistently across all chapters and use the prevalent style manual of the discipline.
- References, tables, and figures should follow the most appropriate style manual of the discipline. Some have the caption centered and set in bold font.
- A footnote should be included if the chapter is co-authored (an example of this is in NDSU guidelines), and including publication information in the footnote or in the Acknowledgments section is recommended.
- The general recommendation about spacing between items and their surrounding non-textual content (equations, figures, tables, quotes, pseudocode, etc.) is to set a consistent spacing between items and their surrounding content, as seen in most academic publications.
- Table (and figure) captions should be in the same font size as general text; however, text inside tables and footnotes may be in smaller font sizes as needed to fit the item within the page margins.
- NDSU guidelines have a number of very specific rules (e.g., Table of Contents and Prefatory List formatting, abstract word count, headings, body text paragraphs, etc.); however, they give a lot of flexibility for what is not covered, and give the student (and committee) control of the written content.
- It should be noted that LaTeX is a vast program with numerous facilities and resources that students can use while developing their thesis/dissertation and improving the document quality. All regular LaTeX commands and features work well with the NDSU thesis class.
- The tables and figures are "floating" items, hence may not appear at the location where they were coded by the users. The LaTeX algorithm will decide the best place to display them, and this behavior can be adjusted/forced by float specifiers combinations (e.g., `[h, t, p, b, !]`). As the floats cannot be broken, they will usually be displayed on the top of the next page. With proper application of these float specifiers and moving the location of the float source code (moving text around), the desired effect can be achieved.
- The `H` placement should be avoided as it is not tagging-compatible, and `[h]` or `[h!]` should be used instead.
- Several of the user-developed shortcut commands utilized in the class can be achieved using the LaTeX direct commands—as the shortcuts are simply `\newcommand` with arguments defined in the class using direct LaTeX commands.
- URLs, especially the wrapping issue at the right edge, should be handled differently, whether they are in the body text or in the bibliography. While the body text is automatically handled by the class, the bibliography URLs are handled by using the penalty commands (refer to "`NDSU-Thesis-Extended.tex`"; link: for details).

14.2 NDSU resources and information for disquisition writers

NDSU website	For more information about:
Graduate School	Programs>Graduate Programs & Deadlines; Graduate Admision>Admitted Students>Checklist & FAQs; Academic Support>Forms & Research and Writing; and About>Center for Writers. (link: Graduate School)
Graduate School > Disquisitions	Deadlines, Formatting guidelines and templates, Submission and publication, and Writing policies and guidelines. (link: Graduate School Disquisitions)
Center for Writers	Writing consultations, writing workshops, disquisition boot camps, writing resources on site and online. (link: Center for Writers)
Formatting Guidelines, Templates, and Forms	Graduate school formatting guidelines, NDSU disquisition templates (MS Word), and Forms. (link: Formatting Guidelines, Templates, and Forms)
Formatting Tutorial Videos—MS Word	Hosted on the GPS Academy YouTube Channel. "Navigating the Review Process" workshop recording and "Word Crash Course" videos on In-Paragraph Formatting; Styles; Section Breaks and Page Numbers; Table of Contents; Tables and Figures; Landscape Pages and More about Tables; Template Overview. (link: Formatting Tutorial Videos—MS Word)
	Links accessed on 18 May 2026

14.3 Limitations of the class

- *Tagged PDF creation*: As noted earlier, not all useful and generally popular packages are tagging-compatible. Thus, the users should check the tagging status and use direct or equivalent compatible package commands to achieve the desired result.
- *Shortcuts for tables*: Unlike the figure environment, tables vary a lot (e.g., number of columns and rows), hence not possible to develop shortcuts (such as `\mytfig{}`). Therefore, no shortcuts were coded in the class, and the users should go for the basic command using the `\table` environment.
- *Citations in captions*: This means using `\citep{...}` or `\citet{...}` or `\cite{...}` inside the `\caption{...}` argument of tables and figures is not supported. However, the output of the cites can be hard-coded (after noting the output outside of the caption) as a workaround. It should be noted that cite commands are known to work in captions of other classes.
- *Limited packages loaded*: Only limited packages (see section 7.2) that are thought to be useful for students and cater to most thesis/dissertations, in general, are included in the class. Therefore, for special needs, users need to include the respective packages in the preamble, and that can be loaded by the users as required.

14.4 Voice of the TEX and LATEX developers!

It is interesting to know what the original developers of TEX and LATEX have to say about this system of document preparation. The following are the quotes from the developers about how people feel, perceive, and use the system for their documentation needs.

> "*I never expected TEX to be the universal thing that people would turn to for the quick-and-dirty stuff. I always thought of it as something that you turned to if you cared enough to send the very best.*" —Donald Knuth (Developer of TEX [on which LATEX is based])

> "*LATEX is easy to use—if you're one of the 2 % of the population who thinks logically and can read an instruction manual. The other 98 % of the population would find it very hard or impossible to use.*" —Leslie Lamport (Developer of LATEX)

It is safe to assume that students who came this far should have "cared enough" to improve the quality of their thesis/dissertation, and some who may think they are in the 98 % might discover that they have better logical skills than they originally believed. Furthermore, using LATEX for the documentation needs (e.g., thesis/dissertation, paper, report, book, letter, CV, and so on) should be considered a useful skill in itself that students can pick up and use throughout their career.

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